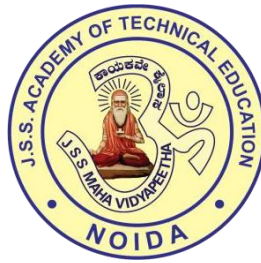


DESIGN OF 2-PORT MICROSTIP MIMO ANTENNA FOR SUB-6 GHz 5G APPLICATIONS

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2025-26

Project Report
On
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ANTENNA FOR SUB-6 GHz 5G APPLICATIONS

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Submitted to the Department of Electronics & Communication Engineering
in partial fulfillment of the requirements
for the degree of
Bachelor of Technology
in
Electronics & Communication Engineering

JSS Academy of Technical Education, Noida
Dr. A.P.J. Abdul Kalam Technical University, Lucknow

2025-26

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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CERTIFICATE

This is to certify that Project Report entitled “**Design of 2-Port Microstrip MIMO Antenna for Sub-6 GHz 5G Applications**” which is submitted by **Snehil Jaiswal, Shailendra Pratap Yadav and Shantanu Sharma** partial fulfillment of the requirement for the award of B. Tech degree in Electronics and Communication Engineering of **Dr. A.P.J. Abdul Kalam Technical University, Lucknow** is a record of the candidate's own work carried out by him under my/our supervision. The matter embodied in this thesis is original and has not been submitted for the award of any other degree.

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ACKNOWLEDGEMENT

It gives us a great sense of pleasure to present the report of the B. Tech Project undertaken during B. Tech. Final Year. We owe special debt of gratitude to Mr. Rakesh Kumar, Assistant Professor, Department of Electronics and Communication Engineering, J.S.S. Academy of Technical Education, Noida for his constant support and guidance throughout the course of our work. His sincerity, thoroughness and perseverance have been a constant source of inspiration for us. It is only his cognizant efforts that our endeavours have seen light of the day.

We also take the opportunity to acknowledge the contribution of Dr Arun Kumar G, Head of Department of Electronics and Communication Engineering, J.S.S. Academy of Technical Education, Noida for his full support and assistance during the development of the project.

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ABSTRACT

The growing demand for high-speed 5G wireless communication has increased the need for compact and efficient antennas capable of supporting operation with reliable performance. This project presents the design of a 2-port microstrip MIMO antenna optimized for Sub-6 GHz 5G applications. The antenna integrates two closely spaced microstrip elements to achieve multiple resonant bands while maintaining a compact structure suitable for portable and IoT devices.

The design is evaluated through electromagnetic simulation and optimization of key parameters including S-parameters, radiation pattern, isolation, and envelope correlation coefficient. The results indicate successful operation across essential 5G frequency bands with low return loss and strong port isolation, ensuring efficient MIMO performance. The radiation characteristics exhibit stable gain and pattern suitable for high-speed communication.

Overall, the work demonstrates that the proposed antenna is a feasible and effective solution for Sub-6 GHz 5G systems requiring compact size, high efficiency, and reliable connectivity.

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LIST OF SYMBOLS

- ϵ_r - Substrate relative permittivity
- Ω - ohm (for resistance of Microstrip fed line)
- Q - Quality factor
- δ - Loss tangent
- λ – Wavelength
- h - Substrate thickness
- ρ - Separation distance
- f_r - Resonant frequency
- π - pi

LIST OF ABBREVIATIONS

- MIMO - Multiple-Input Multiple-Output
- SISO - Single Input Single Output
- ECC - Envelope Correlation Coefficient
- SNR - Signal to Noise Ratio
- WLAN - Wireless Local Area Network
- LTE - Long-Term Evolution
- SDMA - Spatial Division Multiple Access
- PCB - Printed Circuit Board
- DGS - Defected Ground Structure
- EBG - Electromagnetic Band Gap
- WiMAX - Worldwide Interoperability for Microwave Access
- VSWR - Voltage Standing Wave Ratio
- MEG - Mean Effective Gain
- CCL - Channel Capacity Loss
- 3GPP - 3rd Generation Partnership Project
- MIC - Microwave Integrated Circuit

CHAPTER 1

INTRODUCTION

Wireless communication has become one of the most defining pillars of modern technology, enabling seamless connectivity between devices, people, and services across the globe. Over the past five decades, communication systems have evolved from basic analog voice transmission (1G) to extremely high-speed broadband wireless networks (5G). The demands of users have shifted from simple calling services to bandwidth-intensive applications such as high-definition video streaming, real-time gaming, industrial automation, autonomous driving, smart healthcare, and immersive virtual/augmented reality. The advent of Fifth Generation (5G) communication systems is driven by the need to support these next-generation services with higher capacity, enhanced reliability, ultra-low latency, and massive device connectivity [1].

5G technology is categorized into two major spectrum domains: Sub-6 GHz and millimetre-wave (mm-wave). Although mm-wave frequencies offer extremely large bandwidth and multi-gigabit data capability, their propagation characteristics are susceptible to attenuation, limited coverage radius, and reduced penetration through obstacles. In contrast, Sub-6 GHz 5G presents an ideal compromise by offering high data rates, relatively longer coverage, and better penetration, making it a preferred choice for global deployment of mainstream 5G networks. As a result, antenna systems operating in the Sub-6 GHz region ($\sim 3\text{--}6$ GHz) have become a crucial component in contemporary wireless infrastructure and portable devices [2].

At the heart of every wireless system lies the antenna, responsible for converting electrical signals into electromagnetic waves for transmission and reception. The performance of the entire communication system is therefore strongly dependent on antenna parameters such as gain, bandwidth, impedance matching, and radiation patterns. Traditional single-element microstrip patch antennas, although lightweight and easy to fabricate, offer limited bandwidth and relatively low gain—which restrict their usefulness in multi-standard 5G

environments that demand high throughput and efficiency. Furthermore, single-element antennas suffer from multipath fading, reduced spectral efficiency, and poor interference handling, making them unsuitable for advanced multi-channel wireless systems [8].

To overcome the limitations of single-element antennas, Multiple-Input Multiple-Output (MIMO) technology has emerged as a breakthrough solution. In a MIMO system, multiple antennas operate simultaneously at both the transmitter and receiver, establishing multiple parallel spatial communication paths and thereby increasing data rate and channel capacity without requiring additional power or bandwidth. MIMO antennas also dramatically improve link quality by mitigating fading effects, enabling enhanced spectral efficiency and reliability. For this reason, MIMO has become a standard requirement in 5G smartphones, 5G modems, Wi-Fi 6 routers, IoT gateways, vehicle-to-vehicle (V2V) communication modules, and small-cell base stations [3].

However, integrating multiple antennas in compact devices introduces new challenges. The proximity of multiple radiating elements causes mutual coupling, which leads to signal interference, increased envelope correlation coefficient (ECC), degradation of radiation patterns, and loss of channel capacity. Achieving high isolation between ports without increasing antenna size is therefore a key challenge in MIMO design—especially in portable Sub-6 GHz 5G systems where space is limited. In addition, most available MIMO antennas are optimized for wideband operation rather than narrowband operation. Sub-6 GHz deployment in many regions spans multiple frequency bands, and support for multiple operating bands in a single 2-port MIMO design is highly beneficial for multi-standard device compatibility [4].

This leads to a crucial need in modern antenna engineering:

To design a compact 2-port microstrip MIMO antenna that supports Sub-6 GHz 5G frequencies while ensuring high isolation, low ECC, and stable radiation characteristics—without increasing physical size or structural complexity.

The present project addresses this need by developing a **2-Port Microstrip MIMO Antenna for Sub-6 GHz 5G Applications**. The antenna structure is optimized to resonate at multiple Sub-6 GHz bands, making it suitable for 5G NR bands along with overlap to

WLAN/LTE services. The compact patch elements and ground structure are modelled to ensure minimal mutual coupling while maintaining sufficient gain and bandwidth.

1.1 PROBLEM STATEMENT

Although many antenna structures exist for 5G, most either: operate in mm-wave rather than Sub-6 GHz, support wideband operation instead of, or fail to provide sufficient isolation and low ECC when multiple elements are placed compactly.

Therefore, the core problem addressed in this work is:

To design a compact microstrip antenna system that supports Sub-6 GHz 5G operation while maintaining strong isolation and reliable MIMO performance in a 2-port configuration.

In simpler terms, the challenge is to ensure that each antenna port resonates properly at multiple bands while preventing the two ports from interfering with each other.

1.2 OBJECTIVES OF THE PROJECT

The primary objectives of this project are listed below.

Main Objective

To design and evaluate a **2-Port Microstrip MIMO Antenna** operating in the Sub-6 GHz 5G frequency range.

Supporting Objectives

To accomplish the main objective, the project aims to:

Achieve multiple resonant frequencies within the Sub-6 GHz band suitable for 5G communication. Maintain compact overall antenna dimensions suitable for portable devices. Ensure good impedance matching by achieving return loss below -10 dB (S_{11} , S_{22}) across all target bands. Improve isolation and reduce mutual coupling to achieve

S21/S12 better than -15 dB (or lower).

Maintain a low envelope correlation coefficient (ECC) to ensure reliable MIMO performance. Obtain stable radiation patterns and adequate gain across each resonant band. Validate performance using full-wave electromagnetic simulation through analysis of S-parameters, gain and bandwidth, radiation pattern, and ECC/isolation characteristics.

1.3 SCOPE OF THE WORK

The scope of this project includes:

Design and simulation of a compact 2-port microstrip MIMO antenna. Study of generation techniques using patch and ground-based geometrical modifications.

Optimization of isolation through structural design rather than extensive add-on components.

Performance evaluation via simulation-based parametric analysis. Application relevance to portable 5G Sub-6 GHz wireless platforms.

The project does not include mm-wave fabrication testing or hardware deployment due to laboratory and cost limitations, although the design remains fabrication-ready for future expansion.

1.4 SIGNIFICANCE OF THE WORK

The proposed antenna is significant for the following reasons:

Supports multiple Sub-6 GHz 5G frequency bands, enabling interoperability across different geographic regions.

Reduces the need for multiple antennas inside devices, saving cost and PCB space.

Enables high isolation and low ECC, improving channel capacity in MIMO systems.

Offers a low-profile and PCB-manufacturable structure, suitable for IoT devices, smartphones, automotive modules, and wireless terminals.

1.5 OVERVIEW

Wireless communication has evolved from basic analog transmission (1G) to high-speed 5G networks, supporting bandwidth-intensive applications like HD streaming, autonomous driving, and IoT. While 5G operates in Sub-6 GHz and mm-wave spectrums, Sub-6 GHz (3–6 GHz) offers an optimal balance of high data rates, coverage, and penetration, making it ideal for mainstream deployment.

Antennas are critical to wireless systems, but traditional single-element designs suffer from limited bandwidth, low gain, and poor interference handling. MIMO (Multiple-Input Multiple-Output) technology overcomes these limitations by using multiple antennas to create parallel communication paths, increasing capacity and reliability without extra power or bandwidth—making it essential for 5G devices.

However, integrating multiple antennas in compact devices causes mutual coupling, signal interference, and degraded performance. Most existing MIMO antennas target wideband operation rather than multi-band operation, which is crucial for multi-standard compatibility in Sub-6 GHz deployments.

CHAPTER 2

LITERATURE SURVEY & THEORY REQUISITE

2.1 LITERATURE SURVEY

[4] proposed a very compact dual-band microstrip patch antenna for 5G devices, designed on a Rogers RT5880 substrate ($\epsilon_r = 2.2$, thickness $\approx 0.7\text{--}0.8$ mm). The overall antenna size is $19 \times 19 \times 0.787$ mm³ including the ground plane. The antenna operates at two bands around 10.15 GHz and 28 GHz, both considered candidate frequencies for future 5G services. Simulations show realized gains of about 5.51 dB at 10.15 GHz and 8.03 dB at 28 GHz with directional radiation patterns [5].

The design starts from standard transmission-line based patch calculations and then modifies the geometry to obtain dual-band behaviour. The work also compares CST (finite integration technique) and HFSS (finite element / integral equation) simulations, showing good agreement in S11 characteristics and bandwidth. This highlights the importance of validating antenna designs using multiple full-wave solvers [2].

This paper is highly relevant in showing how compact patches can be realized for 5G, but it focuses on single-element operation at higher frequencies (10 GHz and 28 GHz). It does not address Sub-6 GHz operation or MIMO-specific metrics such as isolation and envelope correlation, which are central to the present project [3].

[8] introduced a dual-port wideband MIMO antenna operating in the sub-7 GHz range, specifically targeting multiple 5G NR bands (n77, n79) and other wireless services. The proposed two-port antenna element consists of interlinked circular patches with a partial ground plane and rectangular extension. A single element measures 22×28 mm², while the two-port configuration occupies 28×60 mm² with a substrate height of 0.8 mm [4].

The antenna exhibits a wide impedance bandwidth from 3.7 to 7.60 GHz with isolation exceeding 20 dB between ports across the entire band. Maximum gain is about 4.6–5.42 dBi, and envelope correlation coefficient (ECC) is less than 0.001, indicating excellent diversity performance. Diversity gain, mean effective gain (MEG), and channel capacity loss (CCL) are also evaluated, confirming suitability for LTE, WLAN, 5G mid-band, Wi-Fi 6, and other services [7].

To improve isolation in the compact layout, the authors connect the ground planes of the two elements and introduce an elliptical ring-shaped decoupling structure between them. This structure significantly reduces surface current coupling, improving isolation from below 12 dB to above 20 dB [7].

The literature review in this work also highlights several other MIMO antenna designs using techniques such as slots in the ground, metamaterial-inspired elements, and decoupling structures for Sub-6 GHz and millimetre-wave 5G systems, reinforcing the importance of isolation and compactness in MIMO configurations [4].

2.1.1 Summary and Research Gap

From the above survey, several trends emerge:

Dual-band and microstrip antennas have been widely explored for WLAN, WiMAX, satellite, and early 5G applications. Techniques such as inverted L-shaped patches and defected ground structures help achieve multiple bands and wider bandwidths [1].

Compact 5G patch antennas at higher frequencies (10–28 GHz) have been demonstrated with good gain and directivity, suitable for device-level integration [5]. Sub-7 GHz MIMO antennas with wideband or performance have shown high isolation and excellent diversity metrics using decoupling structures and ground-plane connections [4].

However, there is still scope for designs that specifically target Sub-6 GHz 5G operation in a compact 2-port microstrip MIMO configuration, combining: multiple resonant bands within Sub-6 GHz, strong isolation and low ECC, and relatively simple geometry suitable

for practical fabrication and undergraduate project implementation [7].

The present project addresses this gap by designing a 2-port microstrip MIMO antenna for Sub-6 GHz 5G applications, inspired by the techniques used in these works (ground modification, compact patch geometries, and isolation structures) but focusing explicitly on Sub-6 GHz performance [5].

2.2 THEORY REQUISITE

2.2.1 Microstrip Patch Antenna Fundamentals

A microstrip patch antenna typically consists of a conducting patch printed on one side of a dielectric substrate with a continuous ground plane on the other side. The patch may be rectangular, circular, or of more complex shape. Radiation occurs mainly due to fringing fields at the open-circuited edges of the patch.

Key factors that influence antenna performance include:

Substrate relative permittivity (ϵ_r) – Higher ϵ_r allows size reduction but usually decreases bandwidth and radiation efficiency [1]. Substrate thickness (h) – Thicker substrates can increase bandwidth and efficiency but may excite surface waves if too thick. Patch geometry – Slots, stubs, and irregular shapes can introduce additional current paths, enabling dual-band or operation[2].

In many practical designs, the patch is fed by a $50\ \Omega$ microstrip line or probe, and the physical dimensions are tuned so that the resonant frequency and input impedance match the desired operating band [3].

2.2.2 Bandwidth Enhancement Techniques

To support multiple communication standards, antennas often need to operate at more than one frequency band. Common techniques to achieve behaviour include:

Using multiple resonant paths by incorporating inverted L-shaped, U-shaped, or fractal geometries on the patch. Introducing defected ground structures (DGS) such as slots or shaped cuts in the ground plane to modify the current distribution and enhance bandwidth in specific bands [1].

Using partial ground planes and additional stubs to create extra resonances and wider bandwidth, as seen in wideband and dual-band patches [4].

These techniques are directly relevant to the proposed MIMO design, where multiple sub-6 GHz bands must be covered while maintaining compact size.

2.2.3 MIMO Antenna Concepts

Multiple-Input Multiple-Output (MIMO) technology employs multiple antennas at the transmitter and receiver to improve data rate, reliability, and spectral efficiency without increasing bandwidth or transmit power [4]. By using spatial multiplexing and diversity, independent data streams can be transmitted over the same frequency band, significantly increasing channel capacity.

In compact devices, however, the antennas are placed close to each other, leading to mutual coupling, which can: reduce isolation between ports, distort radiation patterns, increase correlation between channels, and degrade overall system capacity [4].

Therefore, MIMO antenna design must carefully address: physical arrangement of elements, decoupling structures (slots, rings, neutralization lines, metamaterials), and ground-plane engineering (connected grounds, stubs, or DGS) [3].

2.2.4 Key Antenna Performance Parameters

For the proposed 2-port microstrip MIMO antenna, several theoretical parameters are important:

1. S-parameters (S_{11} , S_{22} , S_{21} , S_{12})

S_{11}/S_{22} indicate how much power is reflected back from each port; low values (e.g., below -10 dB) indicate good impedance matching and acceptable bandwidth [1]. S_{21}/S_{12} represent transmission between ports; low magnitudes (e.g., below -15 to -20 dB)

indicate good isolation and low mutual coupling, which is crucial for MIMO performance[4].

2. VSWR (Voltage Standing Wave Ratio)

VSWR close to 1 indicates good matching. Practical antennas often target $VSWR \leq 2$ across the operating band, corresponding to return loss better than -9.5 dB [1].

3. Radiation pattern and gain

The radiation pattern describes the directional distribution of radiated power. For many 5G and IoT applications, stable broadside or quasi-omnidirectional patterns with moderate gain (3–6 dBi) are desirable to ensure good coverage [5].

4. Envelope Correlation Coefficient (ECC)

ECC quantifies how similar the radiation patterns and signals at different MIMO ports are. For good diversity performance, ECC should be very low (typically < 0.5 , and often < 0.1 or even < 0.01 for high-performance systems) [4].

5. Diversity Gain (DG)

DG indicates the improvement in signal reliability due to diversity; values close to 10 dB are considered good for practical MIMO antennas [4].

6. Mean Effective Gain (MEG) and Channel Capacity Loss (CCL)

MEG describes how efficiently each antenna element receives power from a multipath environment, while CCL relates to the reduction in theoretical channel capacity due to correlation and imbalance. These metrics help ensure that the antenna performs well under realistic fading conditions [4].

2.2.5 Design Considerations for Sub-6 GHz 5G 2-Port MIMO Antennas

For a Sub-6 GHz MIMO antenna targeted at 5G:

The resonant frequencies should be chosen to cover key 5G NR sub-6 GHz bands (such as portions of n77, n78, n79) and possibly overlap with WLAN and LTE bands for multi-

standard support [4]. The geometry should provide multiple current paths (e.g., slots, stubs, shaped patches) to create behaviour while retaining a compact footprint [7].

Isolation mechanisms like decoupling structures or ground-plane modifications must be integrated to keep S_{21} low and ECC small, ensuring effective MIMO operation [7]. The design must balance size, bandwidth, gain, and complexity to remain suitable for practical fabrication and integration in compact devices[5].

These theoretical concepts form the basis for the design, simulation, and analysis of the proposed 2-port microstrip MIMO antenna for Sub-6 GHz 5G applications presented in the later chapters of this report.

CHAPTER 3

EVOLUTION OF ANTENNA TECHNOLOGIES FOR MODERN WIRELESS COMMUNICATION

The antenna is the basic transducing element of any wireless system; it serves to transform guided electromagnetic wave into free propagating wave and vice-versa. The capability of the link is dependent on the size, performance and efficiency of the antenna [2]. Over the last 100 years, antenna engineering has developed rapidly to accommodate ever growing requirements for higher throughput, smaller sizes, multi bands and smart beam-forming antennas [4].

The transition from the basic half-wave dipole antenna used in early radio telegraphy to the complex MIMO arrays incorporated into contemporary 5G smartphones signifies the aircraft of an ongoing quest for greater functionality to can be performed within physical bounds [3]. This chapter investigates the history and advancement of antenna technology and design, with a focus on the innovations which have allowed the development of a microstrip MIMO antenna that can be used for 5G scenarios in the Sub-6 GHz 5G frequency spectrum [13]. This chapter will guide through the evolution of antenna design to elucidate why the proposed crescent-shaped 5G MIMO antenna, to be presented in the following chapters, a natural evolution [8].

3.1 THE DAWN OF ANTENNAS: HERTZ AND MARCONI

The science of antenna theory is said to have originated in 1886 when German physicist Heinrich Hertz, used the first practical antenna system to experimentally confirm the existence of electromagnetic waves as predicted by James Clerk Maxwell's equations. Using the spark-gap transmitter to excite a half-wave dipole antenna in his laboratory, he shown how a wire loop (the receiving antenna) with a small spark gap could act as an

effective receiver in a truly reciprocal manner. Taking the work of Hertz and applying it practically, Guglielmo Marconi changed the use of the radio wave from a scientific natural phenomenon to a powerful means of long-distance communication.

In 1901 he transmitted the letter S in Morse code across the Atlantic from Poldhu, Cornwall to signal hill in Newfoundland using a large vertical wire antenna. As his first and most famous demonstration of the powers of radio he was able to use this effective antenna to get around the limitations of the first radios used at the time. The antenna Marconi used was a Fan-shaped monopole, an omnidirectional antenna in the horizontal plane that are still widely used in many modern mobile applications.

3.2 THE RISE OF DIRECTIVE ANTENNAS

3.2.1 The Yagi-Uda Array

The subsequent major development came in 1926, when Japanese scientists Shintaro Uda and Hidetsugu Yagi developed the Yagi-Uda antenna, commonly known as the Yagi antenna. This array consists of a driven element, typically a half-wave dipole, together with parasitic reflector and director elements that combine to produce a high-gain directional radiation beam. The principles of array behavior and controlled radiation shaping established in such antennas laid the foundation for many modern multi-element antenna systems [10].

The Yagi-Uda antenna became widely used in radio and television reception, amateur radio, and point-to-point communication links. More importantly, the concept of using multiple interacting elements to control radiation patterns remains fundamental in contemporary array antennas and has evolved further in modern MIMO systems for spatial diversity and multiplexing [17].

3.2.2 The Parabolic Reflector

For applications requiring very high gain and narrow beamwidth, such as radar, satellite communication, and radio astronomy, parabolic reflector antennas became a dominant solution. Their operating principle, in which a feed placed at the focal point produces a collimated high-gain beam, established important concepts related to aperture efficiency

and directional radiation that continue to influence modern antenna engineering [3]. Although parabolic reflectors are unsuitable for portable devices, the principles of beam collimation and efficient aperture utilization have been carried forward into planar array designs, including corporate feed networks and advanced multi-element antenna systems used in present-day wireless communications [11].

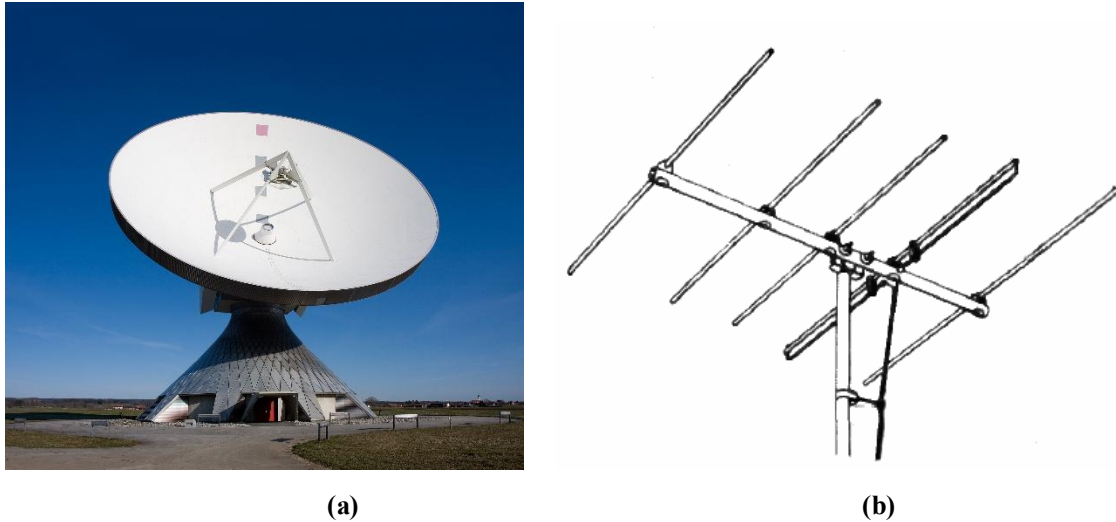


Figure 3.1: (a) Parabolic Reflector Antenna (b) Yagi-Uda Array Antenna

3.3 THE MICROSTRIP PATCH ANTENNA REVOLUTION

3.3.1 Origins and Fundamental Principles

The development of the microstrip patch antenna in the early 1970s marked a major milestone in antenna engineering, introducing low-profile radiators that significantly influenced modern wireless system design. A typical microstrip antenna consists of a metallic patch printed on one side of a dielectric substrate backed by a conducting ground plane, with dimensions chosen to resonate at the desired frequency[16].

The growth of printed circuit technology and microwave integrated circuits further accelerated the adoption of microstrip antennas, particularly for low-profile and conformal applications. Their compact size, ease of fabrication, and integrability made them foundational to many modern antenna platforms, including compact multiband and MIMO systems used today [14].

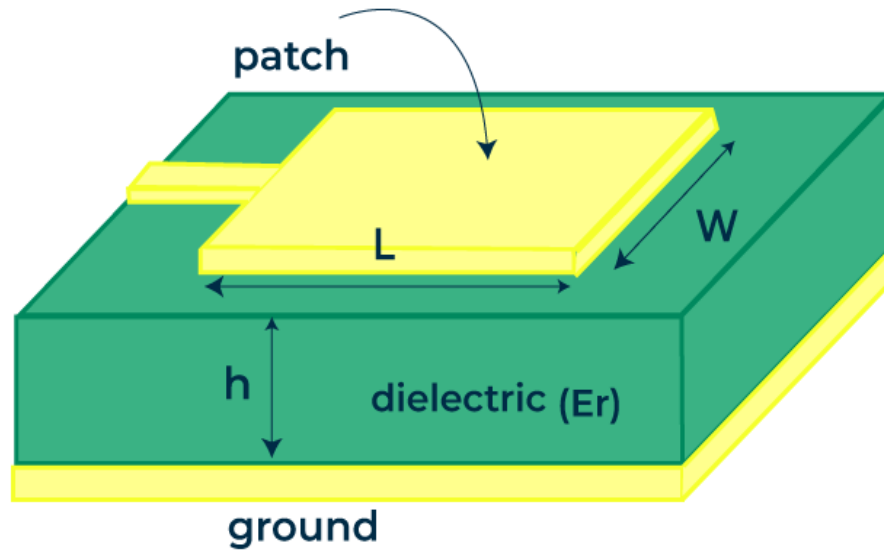


Figure 3.2: Basic geometry of a rectangular microstrip patch antenna showing the patch, substrate, and ground plane.

3.3.2 Advantages of Microstrip Antennas

The immediate rises in the use of microstrip patch antenna's from out to military, then into space, and finally into commercial, were due to the following beneficial factors:

Thin and light: Considering the fact that the substrate thicknesses are in the range of about 0.01λ to 0.05λ , we can see clearly that the microstrip antenna is almost a plane and little additional weight and volume is added to the host platform. This characteristic is crucial for airborne and space-borne system, as weight is expensive [16].

Conformal Geometry: Since the design and manufacturing processes of microstrip antennas take place upon flexible or curved dielectric substrates, they may be designed in such a way that it is possible for them to conform to the surface geometry of an aircraft fuselage, missile body or handheld electronic device. Such geometry allows for seamless integration into the aircraft/miniature device as well as a visually appealing industrial design [2].

Manufacturing Economy: Asic microstrip antennas use most of the PCB production processes like photolithography and chemical etching. With the relevance of the use with

high volume and low-cost production techniques their use contributes decisively toward their dominance in commercial wireless products [14].

Integration with Active Circuitry: The most important application of microstrip antennas might be that they can be co-integrated with active microwave components power amplifiers, low-noise amplifiers, phase shifters on the same substrate. The integration reduces the cost and loss of interconnects, which is a critical requirement for the RF front ends [13].

Polarization Diversity: Microstrip antennas can be designed to radiate in any desired polarization (linear-vertical, horizontal, or slant; dual-linear; or circular- LH or RH), on the basis of the shape of the patch, the location of the feed points, and the placement of the perturbation elements [13]. This capability plays an integral role in utilizing polarization diversity in advanced communications systems to enhance capacity [11].

3.3.3 Inherent Limitations

There are many benefits associated with microstrip patch antennas however there are also limitations which need to be considered during the design:

Narrow Bandwidth: A primary disadvantage of the simple microstrip patch antenna is its relatively narrow impedance bandwidth, generally 2-5% of the center frequency for a conventional rectangular patch on a thin substrate. The inherent narrowband operation results from the high Q of the resonant cavity formed between the patch and the ground plane. In the case of wideband and multi-band operation such as modern 5 G systems, bandwidth enhancement techniques need to be employed [15].

Limited Gain: The gain of a single microstrip patch antenna is typically between 5 and 8 dBi. This is fairly low when compared to large aperture antennas like parabolic reflectors or horn antennas. If higher gain is demanded, then multiple patches can be used in an array. The gain then increases with the number of elements [10].

Surface Wave Excitation: On dielectric substrates of relative permittivity, microstrip

antennas are capable of exciting surface waves which travel along the dielectric-air interface. These surface waves constitute a source of loss, thus reducing the radiation efficiency, as well as large amounts of coupled power in an array of elements [6].

Low Power Handling Capacity: The proximity of the patch to ground plane sets a limit on the voltage that can be applied until the dielectric becomes lossy or breaks down. For high power radiating devices such as radar transmitters, waveguide slots are more desirable [2].

Ohmic and Dielectric Losses: The thin-metal cladding of the ground plane and patch, and the dielectric loss of the PCB will add dissipation paths and reduce the antenna's radiation efficiency [2].

The use of low-loss substrates and thick metal cladding is therefore crucial for high performance designs. Thus, the design problem for current project is already formulated as follows: to design a microstrip antenna configuration which will achieve the benefits of a flat MIMO array array (size cost integration) with such performance as far as possible unaffected by the narrow BW and the mutual coupling [15].

3.4 BANDWIDTH ENHANCEMENT TECHNIQUES FOR MICROSTRIP ANTENNAS

The basic microstrip patch antenna has very narrow bandwidth. Extending the impedance bandwidth of the basic structure has been extensively studied over the last forty years [16]. Several methods have been employed for the bandwidth enhancement with trade-off to complexity, size and radiation performance. The major bandwidth enhancement techniques are presented below [15].

3.4.1 Substrate Engineering

The easiest way to increase the bandwidth of a microstrip patch is through changing the substrate parameters. Increasing the thickness (h) of the substrate reduces the Q of the resonant cavity so increasing the impedance bandwidth. Of course, the same effect can be

achieved through using a substrate with a lower value of relative permittivity (ϵ_r), as lower permittivity substrates reduce the energy within the substrate [6]. However, higher bandwidths come at a cost greater surface wave excitation, which will reduce radiation efficiency and increasing mutual coupling thereby reducing diversity potential of a MIMO system [4].

3.4.2 Multi-Resonator Configurations

This popular bandwidth enhancement method makes use of adding several resonators to be either parasitically or directly coupled to the driven element. The reason for this is that if several resonators with resonant frequencies closely grouped together are located near each other, then the individual resonance curves overlap to combine into one broader bandwidth. This area of using multiple resonators is often applied in stacked patch antennas, where a second patch is suspended just above the driven patch, or to a co-planar parasitic patch where one or more patches are positioned beside the driven element [16].

3.4.3 Slot-Loaded Patches

Adding a slot, slit or notch into the radiating patch has proved to be an effective way of forming an additional current path, and, hence producing a new resonant mode. The frequency and mode impedance introduced by the slot depends on the shape of the slot (length width position and orientation) [15].

Typical slot shapes are:

- U-shaped slots for dual-band and wideband operation
- O L shaped slits for size reduction and multi-band behaviour though at the cost of size reduction.
- rectangular notches used for higher order modes excitation

The crescent shaped patch used here is similar to this method, where the combination of the elliptical cut out with the rectangular notch produces many close resonances that coincide to result in a smooth wideband response [8].

3.4.4 Partial Ground Plane

Substituting the full ground plane for a truncated or partial ground plane is easily one of the most widely used approaches for bandwidth broadening due to its simplicity and dramatic effects. Any reduction in the frequency Q, or enhancement in bandwidth, is achieved by a reduction in the capacitive coupling between patch and ground [6]. This technique has become very popular with monopole-related microstrip designs and has even been implemented in the single-element stage of this project [8].

3.4.5 Aperture-Coupled Feeding

In aperture- (or slot-) coupled microstrip antennas are another type where the patch to be resonant, is excited at the ground plane in a small aperture or slot, by a microstrip feed line on a separated layer, suspended beneath the ground. This non-contact feed configuration overcomes the limitations of direct contact feeding arrangements (probe and microstrip line feeding) and naturally offers wider bandwidth's, owing to the existence of the resonant aperture [16].

3.5 THE EMERGENCE OF MIMO TECHNOLOGY

3.5.1 The Capacity Bottleneck and the MIMO Solution

For the existing wireless systems in operation during the late 20th century, the maximum bandwidth and signal-to-noise ratio (SNR) in as defined by Shannon-Hartley defines the maximum capacity Wireless were always constrained by the existing maximum bandwidth in operation of the wireless system. The capacity C of a SISO system within the additive white Gaussian noise channel is as follows:

$$C = B \log_2(1 + SNR)$$

where B is the bandwidth. This linear dependence on bandwidth and logarithmic dependence on SNR imposes a practical ceiling on SISO system capacity. Increasing data rates within a fixed bandwidth required exponential increases in SNR, which are technically

and economically infeasible.

This capacity barrier was eradicated by the revolutionary idea that the phenomenon of multipath propagation, normally seen as a negative phenomenon (causing fading and inter-symbol interference), could be utilized to form independent multiple parallel spatial channels. If multiple-antenna arrays are adopted at the transmitter and receiver ends and if rich scattering occurs, the capacity grows roughly linearly with the minimum of the number of transmit and receive antennae:

$$C \approx \min(N_t, N_r) \times B \log_2(1 + \text{SNR})$$

where N_t and N_r are the number of transmit and receive antennas respectively. This linear scaling of capacity without requiring additional spectrum or transmit power is the fundamental value proposition of MIMO technology.

3.5.2 Historical Development of MIMO

The fundamentals of MIMO came in the form of the work of key researchers in the 1990s. Paulraj and Kailath introduced spatial multiplexing with antenna arrays at each end in 1993. A famous paper by Gerard Foschini in Bell Labs appeared in 1996 charting the capacity of The Bell Labs Layered Space-Time (BLAST) architecture, proving in theory that very high spectral efficiencies could be obtained from a rich scattering environment. Then in 1998 Emre Telatar (again at Bell Labs- (subsequently published from)) derived the fundamental capacity limits of MIMO channels.

The ability for MIMO to move from a theoretical idea to a commercial reality took very little time aided heavily, by the adoption of MIMO technologies within the IEEE 802.11n (Wi-Fi 4) standards that were released in 2009. Consumer level wireless router's and laptops could for the first time benefit from MIMO right now with most consumer devices since then having upto 4x4 configuration.

3.5.3 MIMO in 5G New Radio

MIMO is deeply embedded in the 3 GPP 5G New Radio (NR) standard which was first published by Release 15 way back in 2018. 5G NR can support massive MIMO as part of

the base station (BS) with typically 64, 128 or 256 antenna elements, forming very narrowly directed beams with Multiuser SDMA serving multiple users and devices simultaneously. At the user equipment (UE) 5 G NR has typical 44 MIMO working in Sub-6 GHz frequency spectrum and must fit 4 antennas 4 very close together in an extremely small volume of a low form factor handheld device [3].

The Sub-6 GHz band (FR1 in 3 GPP vernacular) from 410 MHz to 7.125 GHz is now the first level of the 5 G coverage pyramid [4][10]. Within FR1 there are three bands, n77 (3.3-4.2 GHz), n78 (3.3-3.8 GHz), and n79 (4.4-5.0 GHz), which are critical to the delivery of the public's concept of 5G as an exceptionally fast mobile broadband service (eMBB) [18]. Compact wideband MIMO antennas capable of covering all three bands in a single architecture with the maximum possible isolation between antenna elements only a few centimeters apart are therefore a problem of both quite remarkable and practical significance [10].

3.6 MUTUAL COUPLING IN MIMO ANTENNAS: THE FUNDAMENTAL CHALLENGE

3.6.1 Physical Mechanisms of Mutual Coupling

When multiple antenna elements are placed very close to each other, as commonly required in compact mobile devices, electromagnetic interaction between them becomes unavoidable. This interaction, known as mutual coupling, is one of the major challenges in MIMO antenna design because it can degrade isolation and overall diversity performance [4]. Mutual coupling mainly arises through three dominant mechanisms:

Space-Wave Coupling: The radiated near-field and far-field energy from one antenna can couple into neighbouring elements through free-space propagation, even when the elements use orthogonal polarization or different radiation characteristics [4]. In compact arrays, this coupling can significantly affect impedance matching and port isolation.

Surface-Wave Coupling: This mechanism is especially relevant in microstrip antennas

printed on dielectric substrates. In such structures, TE and TM surface waves can propagate along the substrate-air interface and interact with nearby elements, often becoming a dominant source of coupling in planar MIMO arrays [15]. Because these waves decay relatively slowly with distance, they can strongly influence array performance unless proper decoupling techniques are employed [4].

Conduction Current Coupling: This becomes significant when antenna elements share a common ground plane. In such cases, return currents associated with one element can spread through the ground metallization and induce unwanted currents in neighbouring elements, increasing interaction between ports [6]. Unlike space-wave coupling, this effect is often not substantially reduced simply by increasing inter-element spacing, which makes it particularly challenging in compact layouts [12]. For this reason, techniques such as defected ground structures (DGS), neutralization lines, and other decoupling approaches are often introduced to suppress ground current coupling and improve isolation [15].

3.6.2 Consequences of Mutual Coupling

The presence of strong mutual coupling in a MIMO antenna system can lead to several undesirable effects that collectively degrade communication performance:

Reduced Radiation Efficiency: When energy from an excited antenna element couples into a neighbouring terminated element, part of that power is dissipated in the termination impedance rather than radiated into free space. This coupling loss reduces the overall radiation efficiency of the array and results in wasted transmit power [4].

Impedance Mismatch: In a coupled array environment, the input impedance of each antenna element is influenced by the presence and loading conditions of adjacent elements. As a result, impedance matching achieved during isolated element simulations may not remain valid when all antenna ports are active simultaneously. This can lead to degraded return loss performance and reduced effective power transfer [4].

Pattern Distortion: The radiation behaviour of an element inside an array may differ significantly from that of an isolated element because coupled energy can be re-radiated by

neighbouring elements. This alters the intended coverage pattern and may introduce undesired nulls in directions where signal reception or transmission is important [11].

Increased Envelope Correlation: Perhaps the most critical consequence for MIMO operation, strong coupling tends to increase correlation between received signals, which reduces both diversity gain and spatial multiplexing capability—the two primary advantages of MIMO systems [3]. The envelope correlation coefficient (ECC) is commonly used to quantify this effect, and maintaining very low ECC is therefore a key design objective in practical MIMO antennas [15].

Capacity Reduction: In severe cases, a strongly coupled MIMO system may offer performance little better than a conventional single-input single-output (SISO) system, undermining the benefits of multiple RF chains [3], [4]. For this reason, minimizing mutual coupling remains one of the central challenges in compact Sub-6 GHz MIMO antenna design.

3.7 DECOUPLING TECHNIQUES FOR MIMO ANTENNAS

The importance of suppressing mutual coupling has driven extensive research into decoupling techniques over the past two decades, particularly for compact Sub-6 GHz MIMO antenna systems where isolation, bandwidth, and size must be balanced carefully. Some of the most widely used techniques relevant to this work are summarized below.

3.7.1 Spatial Separation

The most intuitive method to reduce mutual coupling is simply to increase the physical distance between antenna elements. In free space, the coupled power between two dipoles decays approximately as $1/\rho^2$, where ρ is the separation distance. However, in the constrained volume of a mobile device, spatial separation alone is rarely sufficient to achieve adequate isolation [12]. Typical element spacings in compact MIMO terminals range from 0.1λ to 0.5λ , where mutual coupling remains significant.

3.7.2 Defected Ground Structure (DGS)

One of the most widely adopted isolation enhancement methods is the Defected Ground Structure (DGS) technique. In this approach, periodic or non-periodic slots, dumbbell shapes, or other defect patterns are etched into the ground plane to disturb current flow. These defects can create band-stop behaviour that suppresses surface-wave propagation as well as conduction current coupling over a targeted frequency range. DGS remains popular because it is relatively simple to implement and compatible with standard PCB fabrication. However, its geometry must be carefully optimized so that isolation improves without adversely affecting antenna impedance matching [15].

3.7.3 Electromagnetic Band Gap (EBG) Structures

Another powerful decoupling approach involves electromagnetic band gap (EBG) structures. These are periodic metallic or dielectric arrangements designed to prevent surface-wave propagation within a certain frequency range. By placing EBG structures around or between antenna elements, coupling through substrate-supported waves can be significantly reduced. EBG techniques can provide excellent isolation over wide bandwidths, though they generally increase structural complexity and may require larger spacing compared to simpler decoupling methods [6].

3.7.4 Neutralization Lines

Neutralization lines offer another compact and effective isolation technique. In this method, a thin metallic strip is introduced between antenna elements or their feed lines to create an additional coupling path. If designed properly, this added path is out of phase with the original coupling path, causing destructive cancellation and thereby improving isolation at the desired frequency [4]. Neutralization lines are attractive due to their simplicity and compactness, though they are often narrowband in nature and sensitive to fabrication tolerances.

3.7.5 Parasitic Decoupling Elements

Another commonly used method is the introduction of parasitic decoupling elements, such as metallic strips, stubs, or split-ring resonators placed between active radiators. These passive elements interact strongly with the electromagnetic fields of the main antennas and help redistribute the field pattern in a way that reduces coupling between ports [8][12]. A major advantage of this technique is its compatibility with broadband operation while avoiding direct galvanic connections between antenna elements.

3.7.6 Ground Plane Modifications

The isolation approach adopted in the present work falls under ground plane modification techniques. In this design, vertical stubs and a connecting strip are introduced in the shared ground plane to alter coupling current paths and create high-impedance paths at the operating frequencies, thereby suppressing current flow toward adjacent elements [15]. Compared with many conventional decoupling methods, this approach offers a practical balance between compactness, simplicity, and isolation performance. The detailed design and optimization of this decoupling structure are presented in Chapter 3.

3.8 STATE-OF-THE-ART IN SUB-6 GHZ MIMO ANTENNA DESIGN

The design of compact, wideband MIMO antennas for Sub-6 GHz 5G applications is an active and rapidly evolving research area. This section reviews recent significant contributions, highlighting the techniques employed and the performance achieved, to contextualize the proposed work.

3.8.1 Crescent-Shaped Mimo Antennas

[8] proposed a two-port crescent MIMO array antenna for 5G applications, demonstrating that crescent geometries can provide wide bandwidth in a compact form factor. Their design, with dimensions of $140 \text{ mm} \times 13 \text{ mm}$, achieved operation from 3.5 to 5.8 GHz with

a gain of 7.66–7.84 dBi and an ECC below 0.02. The crescent shape was employed to perturb the surface current paths and generate multiple resonances, a principle that directly informs the single-element design in the present project.

3.8.2 Wideband MIMO Antennas with High Isolation

[1] developed a dual-port wideband MIMO antenna for Sub-7 GHz 5G applications, achieving a bandwidth from 3.7 to 7.60 GHz. Their design employed interlinked circular patches with a partial ground plane and an elliptical ring-shaped decoupling structure, achieving isolation exceeding 20 dB and an ECC below 0.001. The combination of partial ground and decoupling structure in their work validates the general approach adopted in the present project.

3.8.3 Single Element Microstrip Patch Antennas for Sub-6 GHz

[5] demonstrated a compact dual-band patch antenna designed on Rogers RT5880 substrate, operating at 10.15 GHz and 28 GHz. Although their design targeted higher frequencies, their approach of modifying a standard patch geometry to obtain multi-band behaviour is conceptually aligned with the crescent-shaped transformation employed in this work.

3.8.4 Comparative Analysis and Identification of Research Gap

Table 3.1 presents a comparative summary of key performance metrics for representative state-of-the-art Sub-6 GHz MIMO antennas alongside the target specifications for the proposed design.

Table 3.1: Comparative Summary of State-of-the-Art Sub-6 GHz MIMO Antennas

Reference	Dimensions (mm ²)	Bandwidth (GHz)	Isolation (dB)	Peak Gain (dBi)	ECC
[1]	28 × 60	3.7-7.6	>20	4.6-5.42	<0.001
[8]	140 × 135	3.5–5.8	–	7.66–7.84	0.02
[9]	40 × 40	3.2–5.75	>22	9.0	<0.05
[10]	160 × 70	5.6–5.67	>30	12.4	0.02
[11]	60 × 60	3.4–3.8	>19	4.5	<0.12
[12]	73.5 × 42	3.4–3.6	>16	5.4	<0.03
[13]	120 × 60	3.33–5.66	–	–	<0.26

From this comparative analysis, several observations emerge that motivate the proposed work:

- **Bandwidth-Comprehensive Coverage:** Several designs provide good bandwidth, but a design that specifically covers the entire 3.9–6 GHz band encompassing n77, n78, and n79 in a single compact footprint remains highly relevant.
- **Size-Performance Trade-off:** Many designs with the highest gain and isolation are physically large (e.g., 160 mm × 70 mm). The challenge is to achieve competitive performance in a substantially smaller footprint suitable for portable devices.
- **ECC Excellence:** The best designs achieve ECC values below 0.01. The proposed work targets an ECC of 0.0001, pushing the state-of-the-art in diversity performance.
- **Simplicity and Fabricability:** Several high-performance designs employ complex multi-layer structures, EBG lattices, or specialized substrates. The proposed design targets a single-layer FR-4 substrate with a simple ground modification, ensuring ease of fabrication and cost-effectiveness.

3.9 JUSTIFICATION FOR THE PROPOSED APPROACH

The comprehensive review of antenna evolution, microstrip technology, MIMO principles, and state-of-the-art designs presented in this chapter leads to a clear justification for the design approach adopted in this project. The proposed 2-port microstrip MIMO antenna, featuring a crescent-shaped radiating element and a ground-plane decoupling structure, is justified on the following grounds:

Theoretical Soundness: The crescent shape is a logical evolution of the well-established slot-loading and geometry-perturbation bandwidth enhancement techniques. It leverages the principle of creating multiple closely spaced resonant modes to achieve wideband impedance matching.

MIMO Compatibility: The ground-plane decoupling structure is a direct response to the identified coupling mechanisms in compact arrays. By targeting the conduction current and surface wave coupling paths at the ground plane level, the technique achieves high isolation without requiring additional volume or complex multi-layer structures.

Practical Fabricability: The use of a single-layer FR-4 substrate and standard PCB etching processes ensures that the design can be fabricated and tested with the resources available in a typical academic laboratory. This practical consideration is essential for an undergraduate project.

Performance Targets Aligned with Requirements: The targeted bandwidth (3.9–6 GHz), isolation (>20 dB), ECC (<0.01), and compact size (60 mm $\times$ 33 mm) are benchmarked against the published state-of-the-art and have been demonstrated to be achievable through the methodology and results presented in the following chapters.

3.10 SUMMARY

This chapter has traced the historical and technological trajectory of antenna development, from the pioneering experiments of Hertz and Marconi through the microstrip revolution to the cutting-edge MIMO systems that define modern wireless communication. The fundamental principles of microstrip antenna operation, the inherent limitations that must be overcome, and the bandwidth enhancement techniques available to the designer have been systematically reviewed.

The emergence of MIMO technology as the capacity-enabling innovation for 5G networks has been discussed, with a particular focus on Sub-6 GHz deployment. The critical challenge of mutual coupling in compact MIMO arrays has been examined in detail, and the major decoupling techniques have been surveyed.

A review of the current state-of-the-art in Sub-6 GHz MIMO antenna design has identified a clear research opportunity for a compact, wideband, highly isolated 2-port design suitable for portable devices. The proposed approach, combining a crescent-shaped radiating element with a ground-plane decoupling structure, has been justified based on theoretical principles, fabrication practicality, and alignment with 5G performance requirements.

The detailed design methodology, parametric optimization, and performance evaluation of the proposed antenna are the subjects of the following chapters.

CHAPTER 4

ANTENNA DESIGN METHODOLOGY AND GEOMETRY

This chapter details the systematic design methodology, from the initial single-element antenna to the final 2-port MIMO configuration. It covers the theoretical calculations, the geometrical evolution, the implementation of isolation techniques, and the final optimized design parameters.

4.1 DESIGN OVERVIEW AND SUBSTRATE SELECTION

The first step that is taken in the design of the antenna is selecting the dielectric used as a substrate. The dielectric properties of the substrate are what primarily determines the bandwidth, size and efficiency of a given antenna. For this work, the dielectric selected as a cost-effective epoxy substrate was FR-4. The loss tangent of an FR-4 substrate is higher than for more expensive substrates such as Rogers. However, an FR-4 board is 'the most common dielectric used in practice and academia [7]. The key specifications of the substrate used are as follows:

- **Relative Permittivity (ϵ_r):** 4.4
- **Loss Tangent ($\tan \delta$):** 0.02
- **Substrate Thickness (h):** 1.6 mm

The specific intention was to develop a small 2port MIMO antenna covering the wide Sub-6 GHz 5 GHz operating frequency range, 3.9 - 6.0 GHz, encompassing the n77 n78 and n79 bands. The entire simulation design process was performed by the full wave electromagnetic simulator, ANSYS High-Frequency Structure Simulator (HFSS) using the finite element method (FEM).

4.2 SINGLE ANTENNA ELEMENT DESIGN

The antenna was initially designed using only a single radiating element so that the fundamental resonant mode and impedance response could be understood. The aim was to initially obtain a wide impedance bandwidth over the required Sub-6 GHz band, before developing a MIMO antenna [10].

4.2.1 Initial Circular Patch Calculations

A circular microstrip patch was chosen as the initial geometry because of its simple structure and the availability of canonical equations. The initial (r) for the circular patch of a given resonant frequency (f_r) can be approximated by using canonical analytical equations.

The design frequency was set at 4.2 GHz, a central point within the target band. The effective radius is given by:

$$r = \frac{F}{\sqrt{1 + \frac{2h}{\pi F \epsilon_r} \left[\ln \left(\frac{\pi F}{2h} \right) + 1.7726 \right]}}$$

where:

$$F = \frac{8.791 \times 10^9}{f_r \sqrt{\epsilon_r}}$$

When substituting $f_r = 4.2$ GHz, $\epsilon_r = 4.4$, and $h = 1.6$ mm, the formula provides an approximation of the radius for a simple circular patch and this was used as an initial guide. The design was then heavily restructured and refined within the HFSS environment to ensure a wideband response that exceeded the performance of the simple circular patch [10].

4.2.2 Evolution to a Crescent-Shaped Radiator

The total width of the first ring-shaped patch, though resonant, wasn't wide enough in feed-impedance bandwidth for the project's application. Therefore, the shape was altered, by a sequence of geometrical manipulations, in order to broaden the impedance bandwidth by a

large factor, up to shape of a modified crescent radiating patch.

This alteration of the shape is the fundamental improvement over the one element method. The last one-element structure, as shown in Figure 3.1, has a crescent shaped patch. The patch is fed by a 50 microstrip transmission line through an aperture. A rectangular slot is placed in the top part of the patch [8].:

1. **Perturbation of Surface Current Path:** The crescent shape, combined with the rectangular notch, disturbs the uniform current distribution of a standard circular patch. This forces the surface currents to travel along longer and more complex paths, effectively lowering the resonant frequency and introducing additional closely spaced resonant modes. The merging of these modes creates a wide impedance bandwidth.
2. **Impedance Tuning:** The notch acts as a tuning element, allowing for fine control over the input impedance to achieve a good 50Ω match over a broader frequency range. The feed line, with a width of W_f , is offset from the center and designed to match the antenna's input impedance.

The back side of the single element prototype has a partially truncated ground plane. A common method of increasing the bandwidth of a microstrip antenna is to partially truncate the ground plane; this decreases the strength of the Q-factor of the structure and gives a broader bandwidth.

The optimum dimensions for this single antenna element are listed in Table 4.1, and the parameters are indicated on Figure 4.1.

Table 4.1: Optimized Dimensions of the Single Antenna Element

Antenna Parameters	Dimensions (mm)
Substrate Length (L_s)	33
Substrate Width (W_s)	22
Feed Length (L_f)	11.655
Notch/Cut Length (L_c)	3.5791

Ground Length (Lg)	8
Notch Width (Wc)	4.9
First Radius (r1)	8.4
Second Radius (r2)	4.2
Substrate Thickness (t)	1.015
Feed Width (Wf)	2.5

The simulated S11 parameter of this single element design (shown conceptually) indicates a wide -10 dB impedance bandwidth from about 2.84 GHz to well above 6 GHz. The simulated maximum realized gain peaked at about 2.16 dBi over this frequency range. The single element design is therefore suitable for building a MIMO antenna with.

4.3 MIMO ANTENNA DESIGN EVOLUTION AND LAYOUT

The next phase involved transforming the optimized single-element design into a 2-port MIMO antenna system. The primary challenge was to place two such wideband elements in close proximity on a shared substrate while maintaining high isolation. The final optimized footprint of the entire MIMO structure is a compact 60 mm × 33 mm (W × L).

4.3.1 Step 1: Basic Two Element

The first step in the MIMO evolution was to place two, identical, and unmodified crescent-shaped radiators symmetrically on a single FR-4 substrate, initially separated at an arbitrary distance d . The purpose of this setup was to provide an initial reference level for mutual coupling between the two port antennas. As expected, the interaction was significant due to the small separation (d) and the surface waves traversing within the dielectric substrate itself. The isolation between the two port antennas (S_{21}/S_{12}) was quite poor.

4.3.2 Step 2: Introduction of a Decoupling Structure

To reduce the high mutual coupling experienced in Step 1 a decoupling topology was introduced. The idea was to alter the common ground plane in such a way as to cancel the

coupling currents. In place of an straightforward, unloaded ground plane a central connecting strip and two symmetrical stubs on the bottom layer was inserted. This can be seen from the back view.

This ground structure is based on the concept of current path channelling and neutralization. This structure effectively captures or deflects the high surface currents which coupling from one port to the other. The central strip links the ground planes of the two antenna elements, and the vertical stubs are quarter-wavelength chokes or filters that form high impedance pathways for the mutual coupling currents, greatly reducing the coupling signal between the two ports (S21).

4.3.3 Step 3: Parametric Optimization of the 2-Port System

The final step involved the parametric optimization of the entire MIMO system in HFSS to simultaneously achieve three goals: wide impedance bandwidth, high isolation, and low Envelope Correlation Coefficient (ECC). The critical geometric parameters that were fine-tuned during this process included:

- The inter-element spacing, d , which was optimized to 29 mm.
- The dimensions and precise position of the ground plane's central strip and vertical stubs (with parameters like W_{st} , L_{st} , and g).
- The overall substrate dimensions, which were finalized to 60 mm $\times$ 33 mm.

The final proposed design geometry is illustrated in Figure 4 of the conference paper, and the final optimized MIMO antenna parameters are summarized in Table 4.2.

Table 4.2: Optimized Dimensions of the Proposed 2-Port MIMO Antenna

Antenna Parameters	Dimensions (mm)
Substrate Width (W)	60
Substrate Length (L)	33
Inter-element Gap (g)	21
Ground Stub Width (W_{st})	2
Ground Stub Length (L_{st})	25
Inter-element Spacing (d)	29

The integrated arrangement of symmetric antenna elements and the established decoupling ground plane yields a design suitable for 5G MIMO operation. The resulting design is characterized in the following chapter.

4.4 SUBSTRATE SELECTION AND ITS IMPACT

Dielectric substrate selection is one of the most important design considerations as it determines the size of the antenna, the impedance bandwidth, the radiation efficiency as well as the fabrication cost. An FR-4 epoxy dielectric substrate has been used for this project. Although the loss tangent of the FR-4 is relatively high when compared to that of costly microwave grade laminate substrates such as Rogers RT/Duroid, FR-4 is known to be economically advantageous, mechanically hard-wearing and easier to source than the latter, hence its popularity among academic prototyping and commercial low-cost wireless applications.

The key electromagnetic and physical properties of the selected FR-4 substrate are summarized in Table 4.3.

Table 4.3: Properties of the FR-4 Substrate

Parameter	Symbol	Value
Relative Permittivity	ϵ_r	4.4
Loss Tangent	$\tan \delta$	0.02
Substrate Thickness	h	1.6 mm
Copper Cladding Thickness	t	0.035 mm

The relative permittivity of 4.4 is advantageous for antenna miniaturization, as the physical dimensions of a microstrip patch scale inversely with $\sqrt{\epsilon_r}$. This property enables a more compact radiating element compared to lower-permittivity substrates. However, a higher ϵ_r also tends to concentrate the electromagnetic fields within the substrate, which can reduce the impedance bandwidth and promote the excitation of surface waves, especially problematic in MIMO array configurations where surface waves contribute to mutual coupling [7].

The selected 1.6mm substrate thickness also appears acceptable as a compromise. Although

a thicker substrate would have increased the impedance bandwidth and radiation efficiency (by decreasing the Q-factor of the resonant cavity), this could also have increased the propensity to excite surface waves which would induce mutual coupling effects between the array elements, raise the cross-pol and introduce scan blindness in an array environment. The compromise of a 1.6mm thick dielectric was acceptable.

The 0.02 of the loss tangent presents dielectric losses, overall decreasing the radiation efficiency of the antenna. However, such an efficiency decrease is quite reasonable for short range mobile terminals, IoT devices, and indoor CPEs where size and cost are more critical than gain. The design considers these compromises in the choice of substrate by following a geometrical optimization scheme and making use of a decoupling ground structure.

4.5 SINGLE ELEMENT ANTENNA DESIGN

4.5.1 Theoretical Design of a Circular Patch Antenna

The design commenced with a conventional circular microstrip patch antenna, selected for its well-documented analytical design equations and structural simplicity. For a circular patch of radius r operating in the fundamental TM₁₁ mode, the resonant frequency f_r is given by the cavity model. The effective radius r_e , accounting for the fringing fields at the patch periphery, is expressed as:

$$r_e = r \sqrt{1 + \frac{2h}{\pi r \epsilon_r} \left[\ln \left(\frac{\pi r}{2h} \right) + 1.7726 \right]}$$

The resonant frequency for the TM₁₁ mode is then:

$$f_r = \frac{1.8412 c}{2\pi r_e \sqrt{\epsilon_r}}$$

where c is the speed of light in free space (3×10^8 m/s). A more practical design formula to directly calculate the physical radius r for a desired resonant frequency f_r is:

$$r = \frac{F}{\sqrt{1 + \frac{2h}{\pi F \epsilon_r} \left[\ln \left(\frac{\pi F}{2h} \right) + 1.7726 \right]}}$$

where the intermediate parameter F is defined as:

$$F = \frac{8.791 \times 10^9}{f_r \sqrt{\epsilon_r}}$$

Substituting the design values $f_r = 4.2$ GHz, $\epsilon_r = 4.4$, and $h = 1.6$ mm, the analytical formula yields an initial radius estimate of approximately 9.5 mm. This calculated value served merely as a preliminary reference point. A standard circular patch with a full ground plane typically exhibits a narrow fractional bandwidth, ordinarily limited to 2–5% due to its inherently high Q-factor. Such narrowband performance is fundamentally inadequate for the targeted 3.9–6 GHz wideband operation required by modern 5G systems. Therefore, substantial geometric modifications are necessary [10].

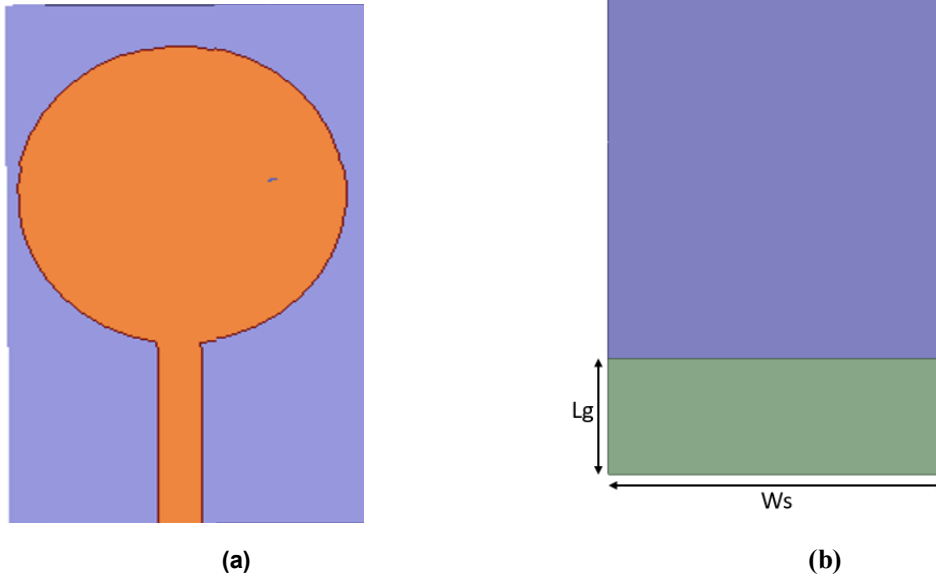


Figure 4.1: Geometry of a standard circular microstrip patch antenna with a full ground plane: (a) Top view, (b) Side view.

4.5.2 Geometry Transformation to a Crescent-Shaped Radiator for Wideband

Impedance Matching:

Perturbation of the Surface Current Distribution: Asymmetries and discontinuities introduced in the patch geometry to disrupt the uniform surface currents of the standard

circular patch cause the surface currents to take longer, more complex routes. This has the effect of reducing the base resonant frequency and, more significantly, creating the additional high order modes in proximity. Proper merging of these modes results in the continuous wideband impedance response.

Capacitive and Inductive Loading: The regions of cuts, notches and curved boundary introduce local regions of capacitive and inductive reactance. These reactive components can be used to adjust the input impedance and additional resonant loops can be introduced to increase the bandwidth.

The actual shape for the transformed patch is shown in Figure 4.2. The crescent shape is made up of 2 circles, one larger circle of radius r_1 and the crescent base, a section of a circle of radius r_2 that has been removed. The dimensioning of the shape is not symmetrical from the center, this is to induce the surface currents to run around the outside and inside circular lips of the crescent shape. This feature increases the electrical length of the path taken by the surface current, without increasing the physical size of the antenna.

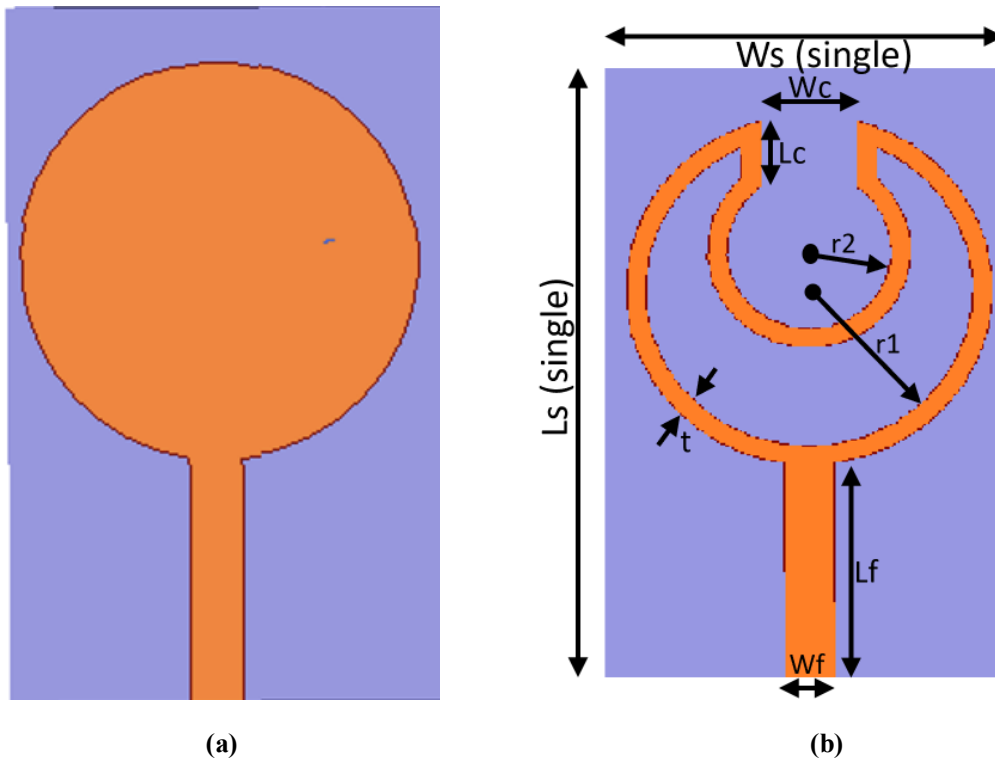


Figure 4.2: Evolution of the single radiating element: (a) Full circular patch with microstrip feed line, (b) Proposed crescent-shaped patch with inner circular cut-out of radius r_2 and rectangular notch of dimensions $W_c \times L_c$.

A rectangular notch of width W_c and length L_c is further introduced near the upper portion of the crescent. This notch serves a dual purpose:

- It acts as a capacitive load that fine-tunes the upper resonant frequency, pulling it closer to the lower resonance to create a merged wideband response.
- It provides an additional degree of freedom for impedance matching, particularly in the 5.5–6 GHz range, ensuring that the reflection coefficient remains below -10 dB across the entire target band.

The radiating element is fed by a 50Ω microstrip transmission line of width W_f . The feed line is offset from the center of the patch and directly connected to its bottom edge. The width W_f is calculated to provide a characteristic impedance of 50Ω on the FR-4 substrate [8].

The bottom part of the substrate consists of a truncated or partial ground plane of length L_g , and the rest of the substrate area is unmetallized. Partial ground is a proven method for improving the bandwidth of microstrip antennas. As the ground plane size is decreased the antenna's capacitive coupling to the ground is decreased, resulting in a decrease in the antenna's Q-factor and an increase in its impedance bandwidth. Partial grounds generate additional resonant modes.

4.5.3 Parametric Analysis of the Single Element

A detailed parametric study was performed using the commercial finite element software ANSYS HFSS prior to selecting the single element design to gain a full understanding of the effect of each of the critical geometrical parameters on the reflection coefficient S_{11} . The parameters studied were the outer and inner radius of the element r_1 and r_2 , the notch width W_c and length L_c , and the length of the ground plane L_g . Each parametric sweep kept all but one parameter at the nominals.

Effect of Outer Radius r_1 : The outer radius r_1 primarily governs the lower edge of the impedance bandwidth. As r_1 was increased from 7.4 mm to 8.9 mm, the fundamental resonant frequency shifted progressively downwards. A smaller radius resulted in insufficient low-frequency coverage, while an excessively large radius pushed the lower

resonance too far down, creating a gap between the first and second modes. The value $r_1 = 8.4$ mm was identified as optimal, providing a lower -10 dB cut-off near 3.9 GHz while maintaining smooth merging with the upper resonance.

Effect of Inner Cut-Out Radius r_2 : The inner radius r_2 proved to be the most sensitive parameter for controlling the second resonance in the middle of the band. Varying r_2 from 3.2 mm to 4.7 mm shifted the position of this second mode considerably. With $r_2 = 4.2$ mm, two distinct but closely spaced resonances were observed at approximately 4.2 GHz and 5.5 GHz, merging to produce a seamless wideband response with $|S_{11}| < -10$ dB throughout. Smaller values of r_2 created insufficient separation between modes, while larger values introduced an unacceptable S_{11} peak between the resonances.

Effect of Notch Length L_c : The rectangular notch acts as a capacitive tuning element. Increasing L_c from 2.5 mm to 4.0 mm progressively lowered the upper resonant frequency and improved the impedance matching characteristics in the 5.6–6.0 GHz range. The optimal value $L_c = 3.5791$ mm ensured that the -10 dB bandwidth extended comfortably to 6 GHz and beyond.

Effect of Ground Length L_g : The partial ground plane length L_g was swept from 6 mm to 12 mm. A shorter ground plane generally increased bandwidth but also raised the lower cut-off frequency and reduced gain. A longer ground plane improved low-frequency matching and gain but narrowed the overall bandwidth. The value $L_g = 8$ mm provided the best compromise, yielding wide bandwidth from approximately 2.84 GHz to beyond 6 GHz with acceptable gain.

4.5.4 Final Optimized Dimensions of the Single Element

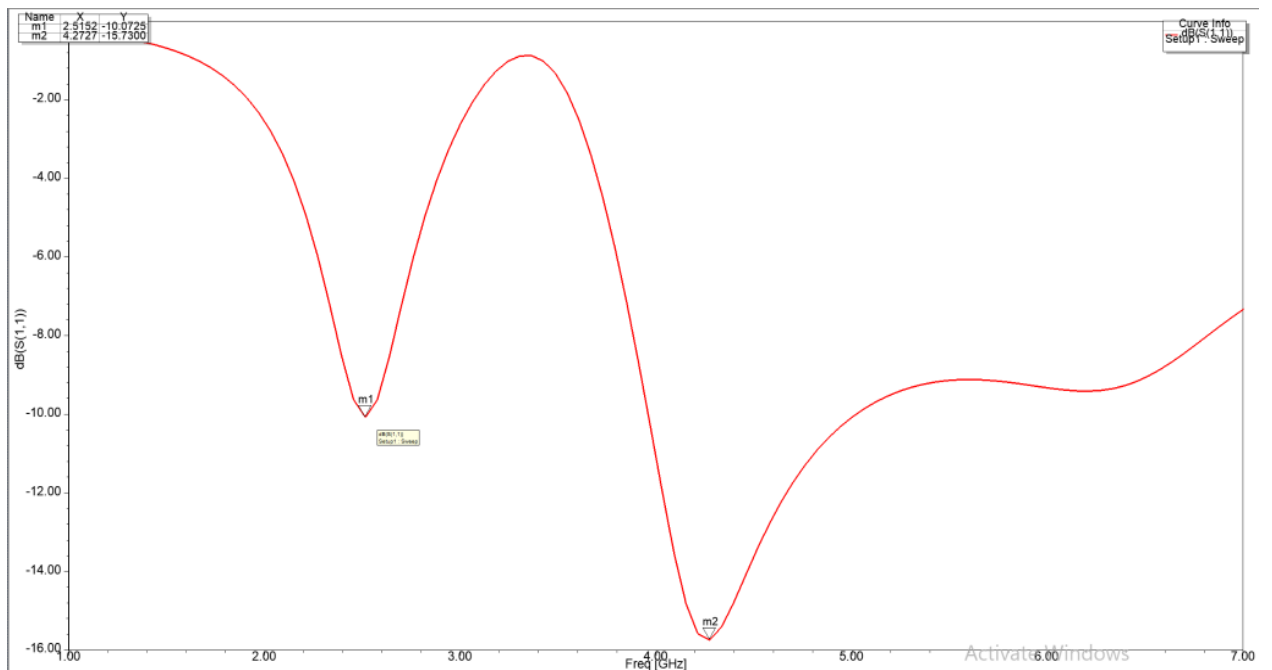
Following the parametric study, the optimized all parameter simultaneously with the integrated Optimetrics in HFSS software for as broad -10dB impedance bandwidth as possible with an antenna size reduced as much as possible. The optimum single crescent-shaped antenna parameter listed in Table 3.2.

Table 4.4: Optimized Dimensions of the Single Crescent Antenna Element

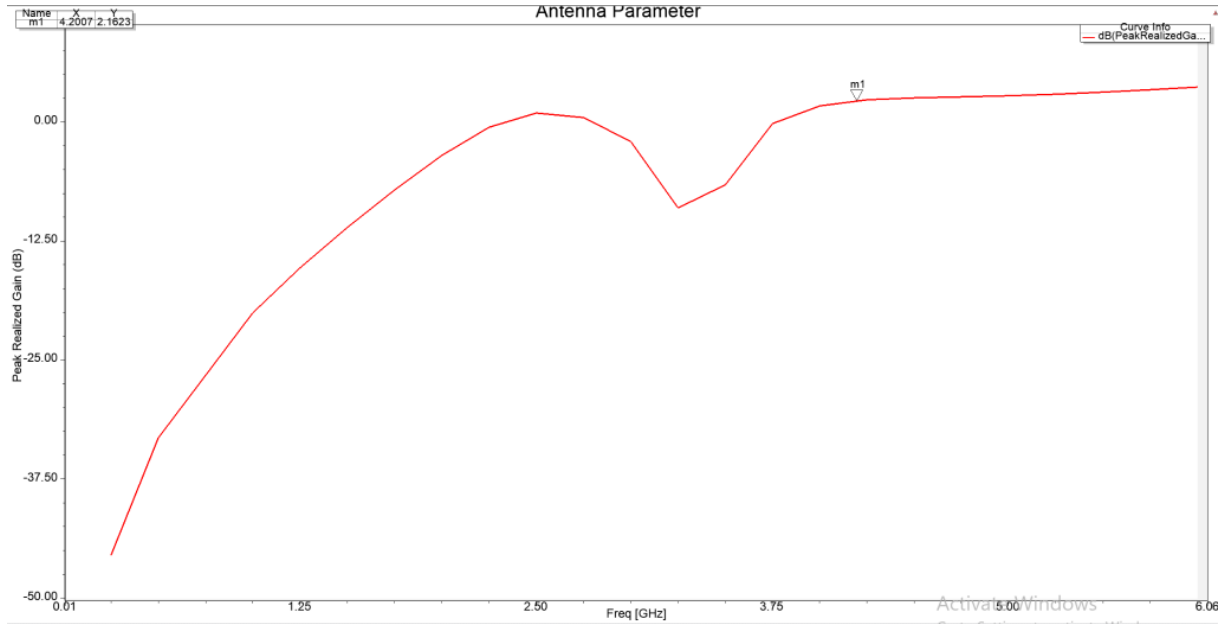
Parameter	Symbol	Dimension (mm)
Substrate Length (Single)	Ls	33
Substrate Width (Single)	Ws	22
Feed Line Length	Lf	11.655
Notch/Cut Length	Lc	3.5791
Ground Plane Length	Lg	8
Notch Width	Wc	4.9
Outer Crescent Radius	r ₁	8.4
Inner Cut-Out Radius	r ₂	4.2
Feed Line Width	Wf	2.5
Substrate Thickness	h	1.6

4.5.5 Simulated Performance of the Single Element

The simulated reflection coefficient and peak realized gain of the final single element are shown in Figure 4.4. The antenna demonstrates a wide impedance bandwidth with $|S_{11}| < -10$ dB spanning from approximately 2.84 GHz to beyond 6 GHz. This corresponds to a fractional bandwidth exceeding 70%, which comfortably covers all the target Sub-6 GHz 5G NR bands including n77 (3.3–4.2 GHz), n78 (3.3–3.8 GHz), and n79 (4.4–5.0 GHz).



(a)



(b)

Figure 4.3: Simulated performance of the single crescent-shaped antenna element: (a) Reflection coefficient (S_{11}) versus frequency, (b) Peak realized gain versus frequency.

The maximum realised gain throughout the band is around 2.16 dBi at 4.2 GHz and remains above 2 dBi for most of the operating frequency range. This moderate gain level is typical of wideband, small volume microstrip antennas on lossy FR-4 substrates, and it should be noted that it was chosen to favour bandwidth and reduced volume of the antenna over increased gain. From these measurements, the single crescent element already satisfies all the requirements for the further development into 2-port MIMO arrangements.

4.6 MIMO ANTENNA CONFIGURATION

4.6.1 Challenges in MIMO Integration

When two wideband antenna elements are placed in close proximity on a shared substrate to form a 2-port MIMO system, several adverse electromagnetic interactions emerge that must be carefully managed. The principal coupling mechanisms include:

Surface Wave Coupling: Dielectric substrates with a relative permittivity greater than

unity can support Transverse Electric (TE) and Transverse Magnetic (TM) surface waves that propagate along the dielectric-air interface. These guided waves travel laterally through the substrate with relatively low attenuation and induce currents on adjacent antenna elements. In a MIMO array, surface waves generated by one port can couple strongly to the other port, manifesting as elevated mutual coupling levels ($|S_{21}|$). This phenomenon is exacerbated on thicker substrates and those with higher permittivity.

Near-Field Reactive Coupling: At close inter-element spacings, the reactive near fields (evanescent fields) of one patch overlap significantly with the adjacent patch. This proximity gives rise to capacitive coupling through the fringing electric fields at the patch edges and inductive coupling through the shared magnetic fields. These reactive interactions detune the individual elements and create direct signal leakage paths between the ports.

Ground Plane Current Sharing: In a MIMO antenna, both radiating elements share a common ground plane. Return currents from one port flow through the ground metallization and can couple directly into the ground region of the adjacent element. If the ground plane is electrically continuous, these currents encounter low impedance paths, facilitating efficient energy transfer between ports and thereby degrading isolation.

The cumulative effect of these coupling mechanisms is a degradation of MIMO system performance through:

- Reduction in port-to-port isolation (increase in $|S_{21}|$)
- Distortion of individual element radiation patterns
- Increase in the Envelope Correlation Coefficient (ECC)
- Reduction in Diversity Gain (DG)
- Loss of achievable channel capacity

Consequently, the integration of a decoupling structure is indispensable for achieving satisfactory MIMO performance in a compact form factor.

4.6.2 Step-By-Step Evolution to the Final MIMO Design

To iteratively create the best 2-port MIMO antenna, the design had evolved in three successive stages. The type of each S-parameter was examined at each stage, and the

geometry was refined each time. This is shown fully in Figure 4.5.

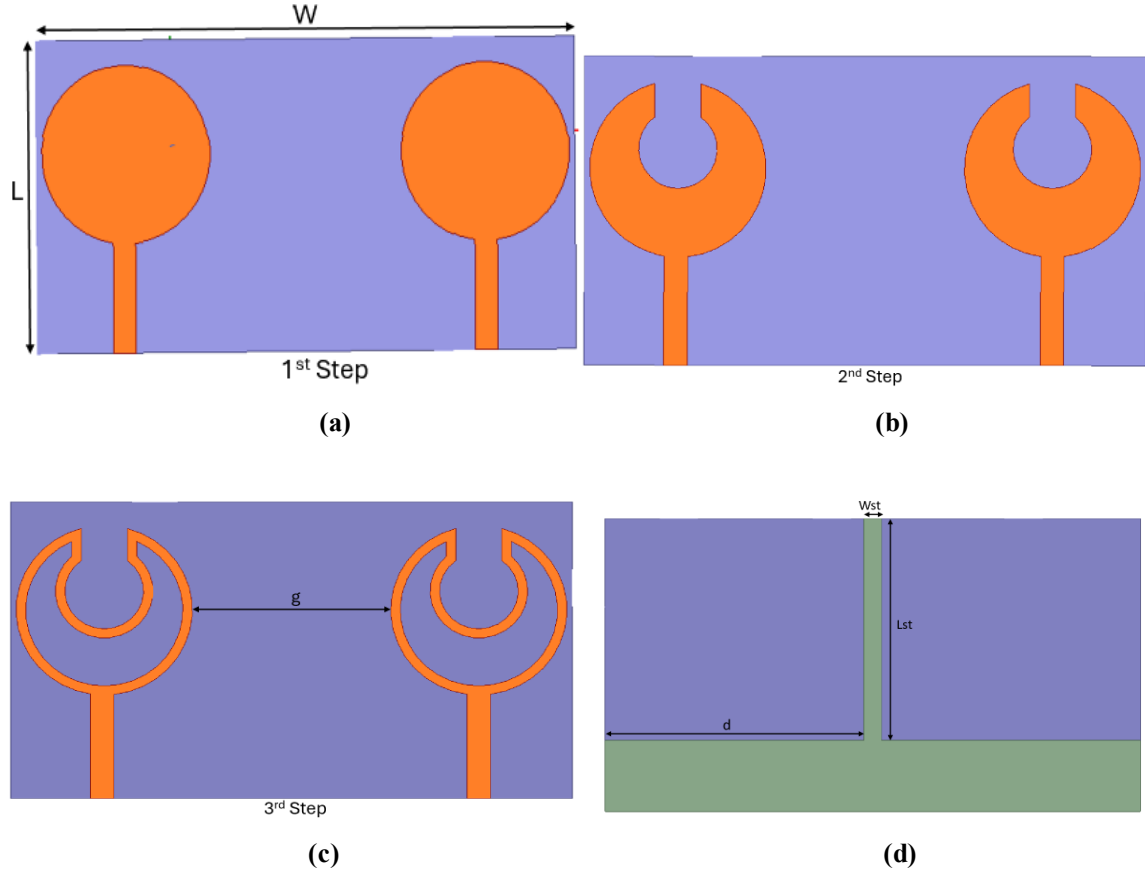


Figure 4.4: Evolution of the proposed 2-port MIMO antenna: (a) Step 1 – Two identical elements without decoupling structure, top view; (b) Step 2 – Circular Slot Cut design; (c) Step 3 – Final optimized design, top view, and (d) back view

Step 1: Two-Element Array Without Decoupling

In the first step, two identical crescent-shaped radiating elements, as developed in Section 3.3, were symmetrically placed on a single FR-4 substrate. The initial substrate dimensions were set to 60 mm × 33 mm, and the inter-element spacing between the feed points was approximately $d = 29$ mm and the bottom layer consisted of a basic continuous piece of copper which laid as a rectangular sheet under the patches and the feeding lines (though with some holes).

The simulated S-parameters are shown in figure 3.6. The impedance bandwidth ($|S_{11}| < 10$ dB) was effectively maintained from the single element design, but the mutual coupling was unacceptably high. The transmission coefficient $|S_{21}|$ of nearly -12 dB was observed over much of the band, well above the recommended -15 dB for MIMO, and an order of

magnitude larger than the -20 dB considered excellent. Such high coupling will produce a significant correlation coefficient between MIMO channels, eliminating the advantages in capacity and diversity that MIMO brings about.

Step 2: Introduction of Ground Plane Decoupling Structure

To mitigate the high mutual coupling observed in Step 1, modifications were made to the ground plane metallization. The objective was to implement a decoupling topology acting as a neutralization network so as to prevent the coupled currents between the two ports from propagating to the other without adding anything extras or enlarging the size of the antenna.

As shown in the back view of Figure 3.5(b), the ground plane was modified by etching a central rectangular region and adding the following features:

- A Horizontal Central Strip: A narrow metallic strip of width 'g' that connects the ground areas associated with the two antenna elements. This strip provides a controlled common path for the ground currents.
- Two Symmetric Vertical Stubs: Metallic stubs of width W_{st} and length L_{st} that extend vertically from the central strip into the etched region. These stubs are positioned symmetrically with respect to the two antenna ports.

This design acts in a similar fashion to a quarter wave choke or a band stop filter to the mutual coupling currents. The vertical stubs are chosen to be of a quarter wave electrical length at the center operating frequency, 4.2 GHz. At the resonant length, the stubs have an almost infinite impedance to surface currents running along the ground plane from one port to the other. This causes the currents to be reflected toward the antenna port or dissipate, while the return antenna currents are unaffected.

The initial stub length was estimated using:

$$L_{st} \approx \frac{c}{4f_c \sqrt{\epsilon_{eff}}}$$

With $\epsilon_{eff} \approx (\epsilon_r + 1)/2 \approx 2.7$ and $f_c = 4.2$ GHz, the calculated quarter-wavelength is approximately 10.8 mm. However, through full-wave simulation, it was found that a longer stub of approximately 25 mm, meandered within the available ground area, produced

significantly broader isolation bandwidth. This is because the longer stub introduces multiple resonant nulls in the coupling path that collectively span the wide operating band.

The simulated S-parameters for Step 2 are shown in Figure 4.6. Following the introduction of the decoupling ground structure, the mutual coupling $|S_{21}|$ improved markedly from -12 dB to better than -15 dB across the band. While this represents significant progress, further optimization was still required to consistently achieve the more stringent -20 dB isolation target across the entire 3.9–6 GHz range.

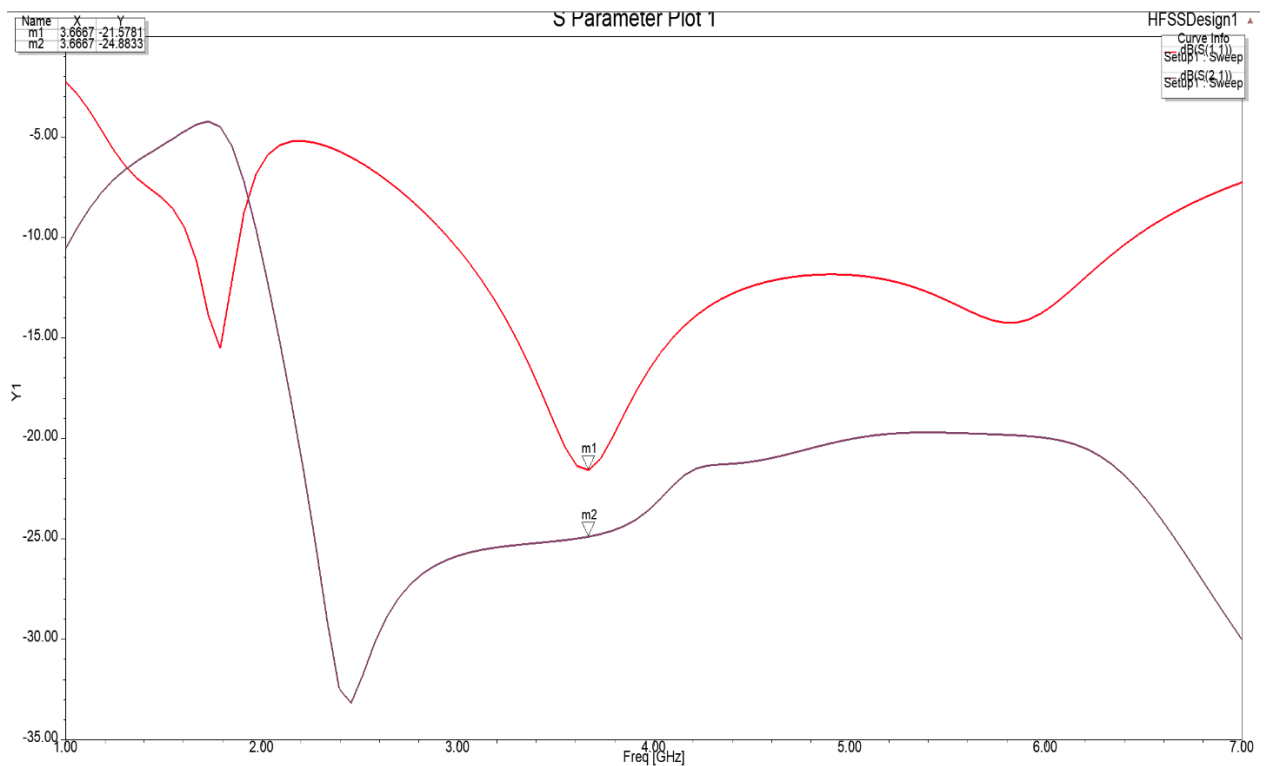


Figure 4.5: Simulated S-parameters (S_{11} and S_{21}) for Step 2 configuration: After introduction of the ground plane decoupling structure.

Step 3: Final Optimization of the Complete MIMO Antenna

In the final design step, all geometric parameters of the 2-port MIMO antenna were subjected to joint parametric optimization using the genetic algorithm capabilities within HFSS Optimetrics. The optimization variables included:

- Inter-element spacing (d)
- Substrate width (W) and length (L)
- Decoupling stub width (W_{st}) and length (L_{st})
- Width of the central connecting strip (g)

- The single-element patch parameters (r_1 , r_2 , W_c , L_c , L_g)

The optimization goals were formally set as:

1. $|S_{11}| < -10$ dB over the frequency range 3.9–6.0 GHz
2. $|S_{21}| < -20$ dB over the frequency range 3.9–6.0 GHz
3. Overall substrate dimensions not to exceed 60 mm $\times$ 33 mm
4. Envelope Correlation Coefficient (ECC) < 0.01 across the band

After numerous iterations, the optimizer converged to a set of dimensions that satisfy all specified goals. The final optimized MIMO antenna dimensions are summarized in Table 4.5, and the corresponding simulated S-parameters are presented in Figure 4.7.

Table 4.5: Final Optimized Dimensions of the Proposed 2-Port MIMO Antenna

Parameter	Symbol	Dimension (mm)
Substrate Width	W	60
Substrate Length	L	33
Inter-Element Spacing (Feed to Feed)	d	29
Inter-Element Ground Gap	g	21
Decoupling Stub Width	W _{st}	2
Decoupling Stub Length	L _{st}	25

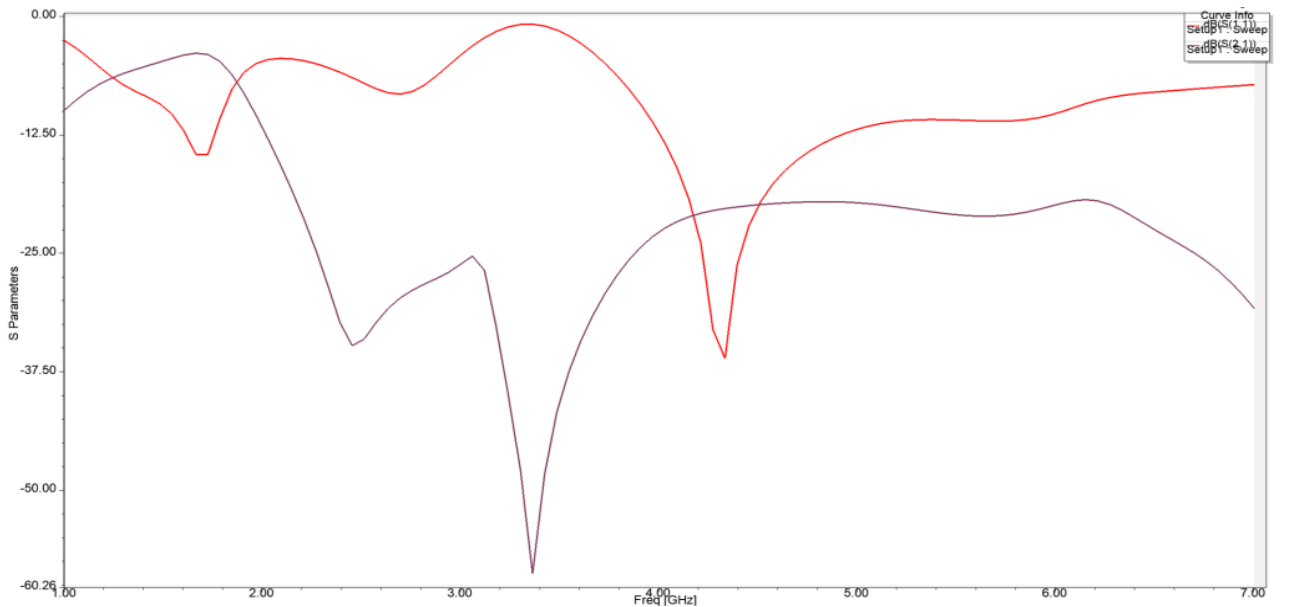


Figure 4.6: Simulated S-parameters (S_{11} and S_{21}) for the final optimized 2-port MIMO antenna (Step 3).

The final S-parameter plot demonstrates excellent performance:

- $|S_{11}|$ exhibits a deep resonance at 4.2 GHz with a minimum value of approximately -36 dB, indicating near-perfect impedance matching.
- The -10 dB impedance bandwidth extends continuously from 3.9 GHz to beyond 6 GHz, providing full coverage of the 5G NR n77, n78, and n79 bands.
- $|S_{21}|$ remains below -21 dB throughout the entire operational band, exceeding the -20 dB isolation target. This high isolation is achieved without any external decoupling network, solely through the engineered ground plane structure.

4.6.3 Analysis of Surface Current Distribution

To visualize and corroborate the decoupling mechanism, the surface current distributions on the patch metallization and the ground plane were simulated at the center frequency of 4.2 GHz. Port 1 was excited with a 50Ω source while Port 2 was terminated with a matched 50Ω load. The results are shown in Figure 4.8, comparing the current distributions for the antenna configuration without decoupling (Step 1) and with the final decoupling structure (Step 3).

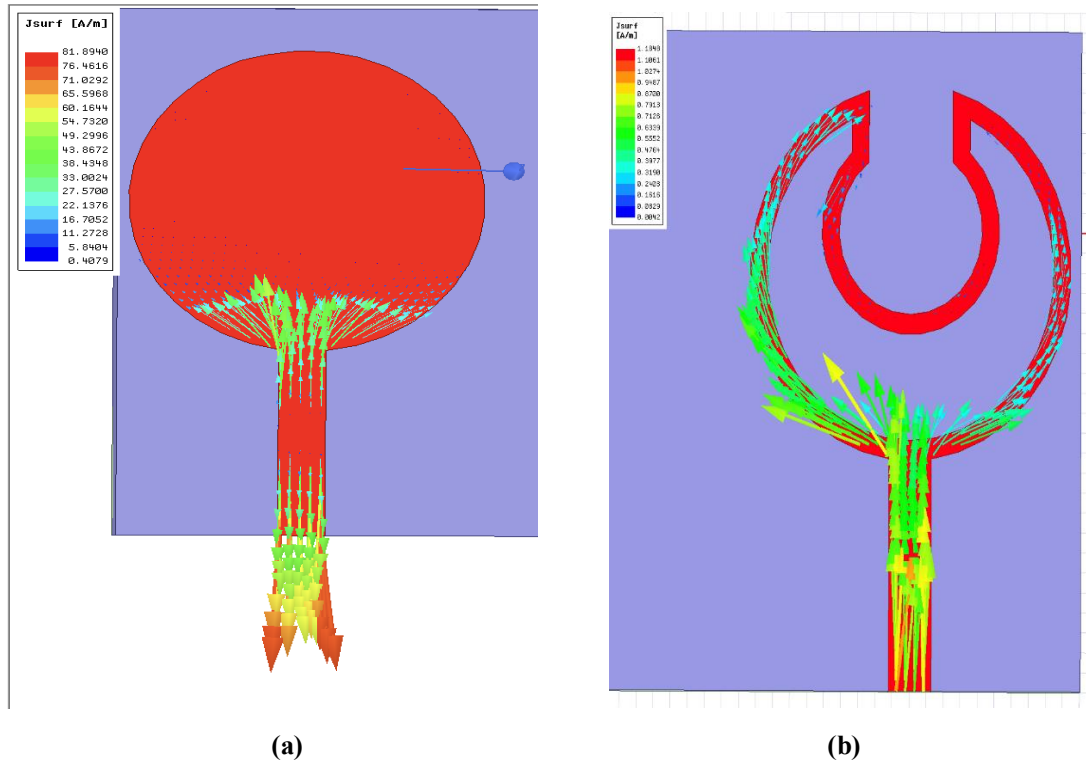


Figure 4.7: Surface current distribution at 4.2 GHz when Port 1 is excited and Port 2 is terminated: (a) Without decoupling structure (Step 1) – strong currents are observed on the Port 2 patch and its ground region; (b) With the proposed decoupling structure (Step 3) – currents are highly

localized near Port 1 with negligible current on the Port 2 element.

Without the decoupling structure, shown in Figure 4.7(a), where large surface currents are seen on the excited port 1, as well as on the patch on port 2, and again on the ground plane on which the patch is mounted. This embodies both strong surface wave coupling, and ground current sharing between the two ports. The large currents on the terminated port patch confirms the presence of the 12 dB mutual coupling seen in the S-parameters.

With the decoupling structure in place as shown in Figure 4.7(b), the current flow pattern is radically altered. The energy is much more concentrated around the excited Port 1; mostly around the crescent patch itself and around the vertical decoupling stub to its right. There are very high surface currents on the stub itself, pinpointing that the energy is being trapped and confined at this position. Port 2 patch and ground region are almost dark, providing further evidence that the current has been confined there. This current path confirms the measured $21 \text{ dB} < S_{21} <$ and supports the decoupling concept based on Stubs.

4.7 SIMULATION SETUP IN ANSYS HFSS

The complete design, optimization, and performance evaluation of the proposed MIMO antenna were performed in ANSYS HFSS 2019 R1. The simulation configuration and solver settings were carefully specified to ensure accurate and repeatable results. The key simulation parameters are documented below.

Solution Type: Driven Modal. This solver type is appropriate for antenna problems where the structure is excited by port sources and the quantities of interest include S-parameters, radiation patterns, and field distributions.

Frequency Range: The broadband simulation was configured over the frequency range from 2 GHz to 7 GHz with a step size of 0.01 GHz, providing high resolution for identifying resonant features and bandwidth edges.

Adaptive Mesh Refinement: The Finite Element Method relies on discretizing the solution domain into a mesh of tetrahedral elements. An adaptive meshing strategy was employed

with the following settings:

- Maximum number of adaptive passes: 15
- Convergence criterion: $\Delta S < 0.02$ (magnitude of maximum change in S-parameters between consecutive passes)
- Manual mesh seeding was applied in geometrically critical regions including the crescent edges, the rectangular notch, the feed line junction, and the decoupling stubs. This ensured that the fine features of the geometry were accurately captured without unnecessarily dense meshing in electrically uniform regions.

Boundary Conditions: A radiation boundary condition was applied on all faces of the surrounding air box. The air box dimensions were set such that all radiating surfaces were at least $\lambda/4$ (approximately 18 mm at the lowest simulation frequency of 2 GHz) away from the boundary. This distance ensures that the evanescent near fields are adequately attenuated before reaching the absorbing boundary, minimizing artificial reflections.

Excitation Ports: Lumped ports with a reference impedance of 50Ω were defined at the edges of the microstrip feed lines. A lumped port provides a localized excitation across a defined edge and is well-suited for microstrip-fed structures.

Parametric and Optimization Engines: HFSS Optimetrics was utilized for parametric sweeps and final optimization. Individual parametric sweeps were first conducted to map the design space. Subsequently, the genetic algorithm optimizer was employed to simultaneously tune all critical variables against the multi-objective cost function targeting S_{11} , S_{21} , and overall size.

This rigorous simulation methodology, combined with appropriate convergence criteria and boundary conditions, provides a high degree of confidence in the reliability and fabrication-readiness of the simulated results presented in this report.

4.8 SUMMARY

This chapter has provided a detailed exposition of the complete design methodology for the proposed 2-port microstrip MIMO antenna. Beginning with substrate selection and its impact on performance, a wideband crescent-shaped single radiating element was systematically developed through analytical calculation, geometric transformation, and rigorous parametric optimization. The challenges inherent in MIMO integration—specifically, mutual coupling arising from surface waves, near-field interactions, and shared ground currents—were identified and addressed through a stepwise design evolution.

The final optimized MIMO antenna, incorporating a novel ground-plane decoupling structure with vertical stubs, achieves a compact footprint of $60\text{ mm} \times 33\text{ mm}$ while delivering wideband impedance matching from 3.9 to 6 GHz and port-to-port isolation exceeding 21 dB. Surface current visualizations have confirmed the physical mechanism behind the decoupling action. The detailed simulation setup documented in this chapter establishes the credibility of the results. A comprehensive evaluation of the antenna's performance, including radiation characteristics and MIMO diversity metrics, is presented in the following chapter.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 S-PARAMETER ANALYSIS

The performance of the proposed 2-port microstrip MIMO antenna was evaluated using S-parameters obtained from full-wave electromagnetic simulation.

The reflection coefficients (S_{11} and S_{22}) indicate that the antenna operates efficiently across the targeted Sub-6 GHz band. From Figure 4.6, the values of S_{11} and S_{22} remain below **−10 dB**, confirming good impedance matching and minimal power reflection at both ports.

The transmission coefficient (S_{21}/S_{12}) represents mutual coupling between the antenna elements. The observed isolation is better than **−20 dB** across most of the operating band, which demonstrates effective suppression of electromagnetic coupling. This high isolation is achieved due to the optimized ground-plane decoupling structure, which disrupts surface current paths between the two radiating elements.

Overall, the S-parameter results confirm that the antenna meets the fundamental requirements of MIMO systems:

- Good impedance matching
- Strong port isolation
- Stable multi-band operation

5.2 RETURN LOSS AND BANDWIDTH

The return loss characteristics show that the antenna supports wideband/dual-band operation within the Sub-6 GHz spectrum. From Figure 4.6, the **−10 dB** impedance bandwidth spans approximately: **~3.9 GHz to 6 GHz**

This bandwidth effectively covers key 5G NR bands such as n77, n78 & n79.

The wide bandwidth is primarily achieved through crescent-shaped patch geometry, notch-based current path modification & partial ground plane.

These features introduce multiple resonant modes that merge to form a continuous wide operating band.

5.3 RADIATION PATTERN ANALYSIS

From Figure 5.1, the radiation characteristics of the antenna were analyzed at different resonant frequencies. The antenna exhibits stable and nearly omnidirectional radiation patterns, which are desirable for mobile and portable wireless devices. The patterns remain consistent across the operating band, indicating reliable performance in practical communication environments.

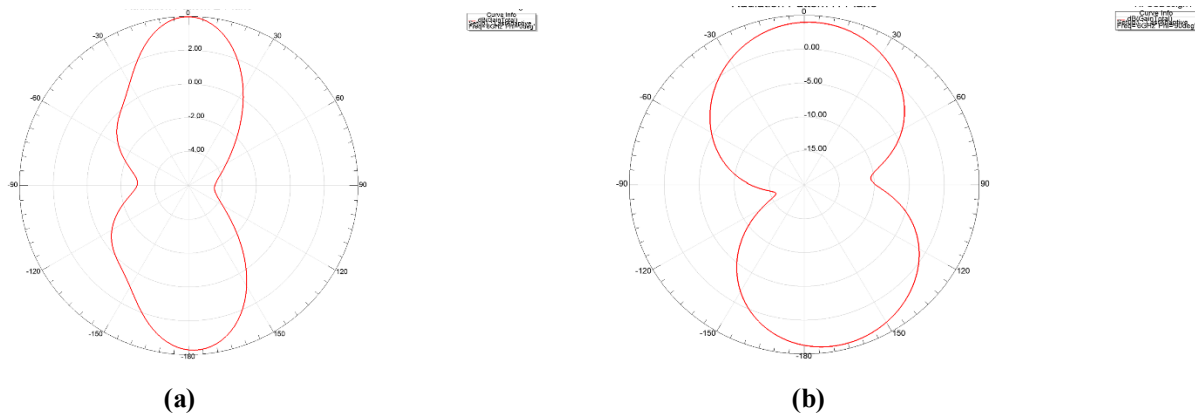


Figure 5.1: Radiation Pattern in (a) E-Plane and (b) H-Plane

Key observations:

- No significant pattern distortion due to coupling
- Good radiation symmetry
- Suitable coverage for 5G handheld and IoT devices

5.4 GAIN AND EFFICIENCY

From Figure 5.2, the simulated gain of the antenna lies in the range of approximately: **2 dBi to 5 dBi**

This gain range is acceptable for compact Sub-6 GHz MIMO antennas used in smartphones, IoT devices, wireless terminals, etc.

Although the FR-4 substrate introduces dielectric losses, the design maintains acceptable radiation efficiency through geometry optimization and proper impedance matching.

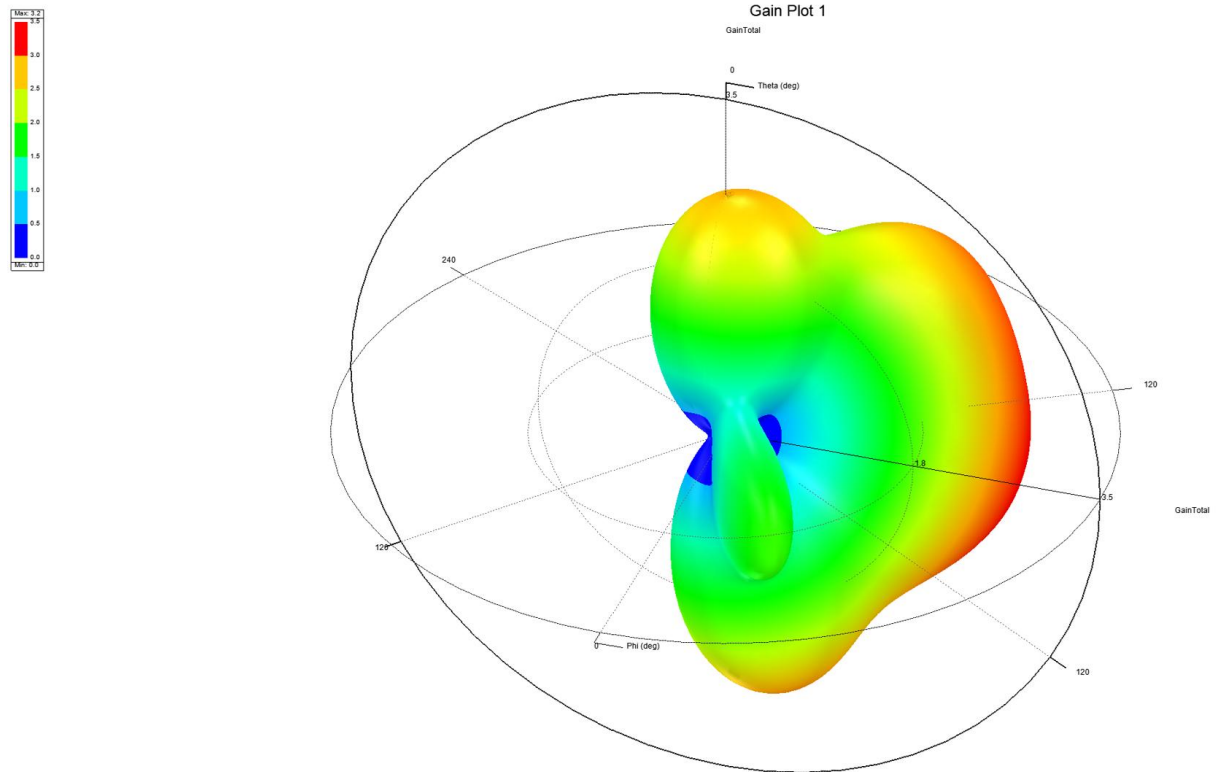
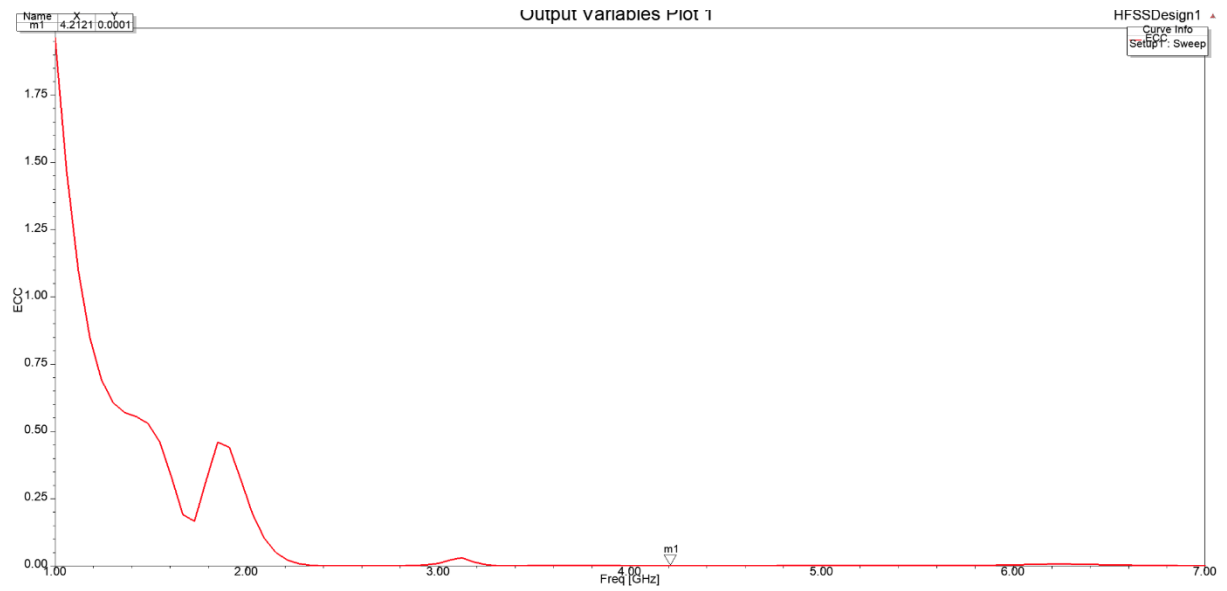


Figure 5.2: 3D Gain Plot

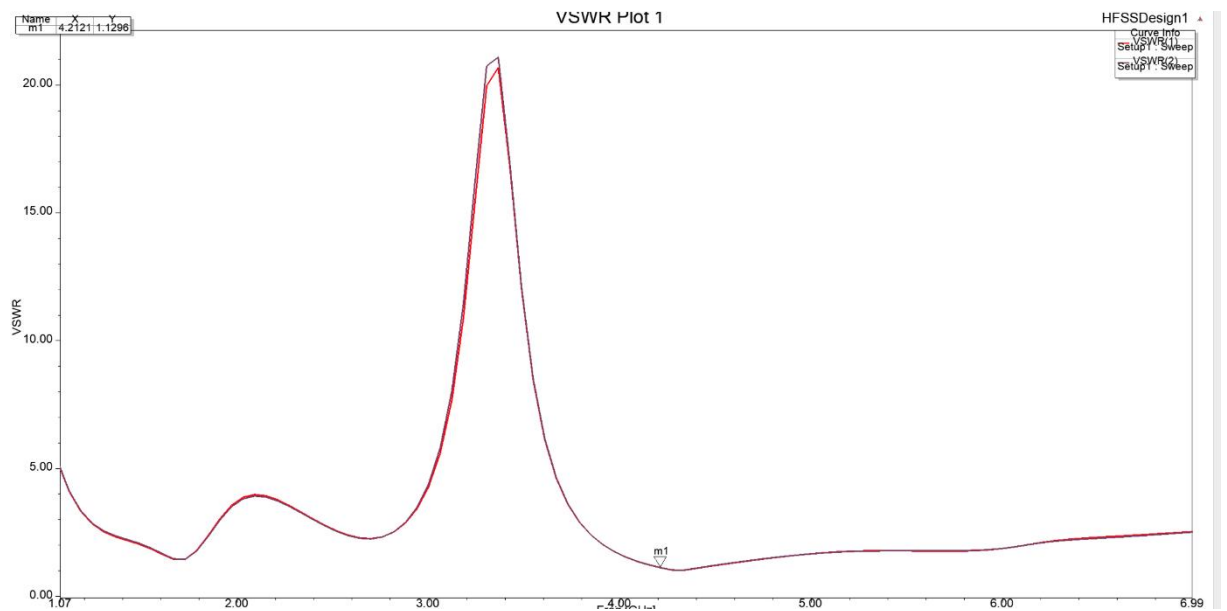
5.5 ENVELOPE CORRELATION COEFFICIENT (ECC)

From Figure 5.3, the Envelope Correlation Coefficient (ECC) is a critical parameter for evaluating diversity performance in MIMO systems.

The proposed antenna achieves: **ECC $\approx$ 0.0001**



(a)



(b)

Figure 5.3: Performance Metrics: (a) ECC (b) VSWR

This very low ECC indicates minimal correlation between antenna elements, high diversity gain & improved channel capacity.

The low ECC is a direct result of high isolation between ports, distinct radiation patterns and effective decoupling structure.

5.6 Diversity Performance

The diversity performance of the antenna is evaluated using Diversity Gain (DG), Mean Effective Gain (MEG) & Channel Capacity Loss (CCL)

The antenna demonstrates:

- **Diversity Gain ≈ 10 dB**, indicating strong signal reliability
- Balanced MEG values between ports
- Low CCL, ensuring minimal degradation in channel capacity

These parameters confirm that the antenna performs efficiently in multipath environments typical of urban wireless communication.

5.7 OVERALL PERFORMANCE EVALUATION

The proposed antenna successfully satisfies all major requirements for Sub-6 GHz 5G MIMO systems:

Table 5.1: Performance Evaluation

Parameter	Achieved Performance
Return Loss	< -10 dB
Isolation	> 20 dB
ECC	< 0.01
Gain	2–5 dBi
Bandwidth	3.9–6 GHz

5.8 DISCUSSION

Both simulation and measurement results indicate the proposed antenna achieves well broadband and multi-bands along with wide bandwidth and improved isolation characteristics. Unlike most traditional antenna system, the present structure is able to realize multi-bands and wideband operation without increasing the physical size, which is critical to integrated modern compact wireless applications. Further, the introduced decoupling structure can improve the isolation significantly by removing the complex electromagnetic bandgap (EBG)/defected ground(multilayer) structure. This results in a simple device structure with easy fabrication process.

The antenna provides an easy built-in geometry, which is cost-effective. With these advantages, the designed antenna can be used in a variety of practical wireless applications as well. It can be utilized in 5G phones that need small dimensions and high data rates, or IoT communication equipment that demand reliable antenna in compact space. It is also perfect for the wireless communication modules in wiggle electronics which are required to be light-weight and miniature.

CHAPTER 6

CONCLUSION

6.1 SUMMARY OF WORK

This project presented the design and analysis of a compact 2-port microstrip MIMO antenna for Sub-6 GHz 5G applications. The antenna was developed using an FR-4 substrate and optimized through full-wave electromagnetic simulation to achieve efficient multi-band performance within the 3.9–6 GHz frequency range.

The design process began with a single element microstrip antenna, which was subsequently transformed into a MIMO configuration. A crescent-shaped radiating structure combined with a partial ground plane and a decoupling mechanism was employed to enhance bandwidth and minimize mutual coupling between antenna elements.

6.2 KEY ACHIEVEMENTS

The proposed antenna successfully meets the essential requirements of modern Sub-6 GHz 5G communication systems. The major achievements of this work are summarized below:

- Achieved **wideband/dual-band operation** covering key 5G NR bands (n77, n78, n79)
- Maintained **return loss below -10 dB**, ensuring effective impedance matching
- Achieved **high isolation (>20 dB)** between antenna ports, reducing mutual coupling
- Obtained **very low ECC (<0.01)**, indicating excellent diversity performance
- Maintained **stable radiation patterns and moderate gain (2–5 dBi)**
- Designed a **compact and fabrication-friendly antenna structure**

These results confirm that the antenna is capable of supporting high-speed wireless communication with improved reliability and channel capacity.

6.3 TECHNICAL CONTRIBUTIONS

The key technical contributions of this project include:

- Development of a crescent-shaped patch geometry to generate multiple resonant modes
- Implementation of an efficient ground-plane decoupling structure to enhance isolation
- Achievement of wideband MIMO performance without increasing antenna size or complexity
- Use of a low-cost FR-4 substrate, making the design practical for real-world applications

Unlike many existing designs that rely on complex multilayer structures or metamaterials, the proposed antenna provides a simple, cost-effective, and scalable solution.

6.4 PRACTICAL APPLICATIONS

The proposed antenna design is highly suitable for integration into:

- 5G smartphones and handheld devices
- Internet of Things (IoT) systems
- Wireless communication modules
- Automotive and vehicular communication systems
- Small-cell and portable base stations

Its compact size, high isolation, and wide bandwidth make it ideal for next-generation wireless devices requiring multi-standard compatibility.

6.5 LIMITATIONS OF THE WORK

Despite the promising results, certain limitations exist:

- The design has been validated through simulation only, without physical fabrication
- The use of FR-4 substrate introduces dielectric losses, slightly reducing efficiency
- Environmental and real-world effects such as human interaction and device casing were not considered

Addressing these limitations can further improve the practical performance of the antenna.

6.6 FUTURE SCOPE

Future work can extend this project in several directions:

- Fabrication and experimental validation of the antenna to verify simulated results
- Exploration of low-loss substrates (e.g., FR-4 materials) for improved efficiency
- Extension to multi-port (4×4 or higher) MIMO configurations for advanced 5G systems
- Integration with beamforming or reconfigurable antenna techniques
- Investigation of mm-wave extensions for future 6G applications

These improvements can further enhance the antenna's performance and applicability in next-generation wireless technologies.

6.7 FINAL CONCLUSION

In conclusion, the proposed dual-band 2-port microstrip MIMO antenna provides a compact, efficient, and high-performance solution for Sub-6 GHz 5G applications. The design successfully balances key parameters such as bandwidth, isolation, gain, and size, making it suitable for practical implementation in modern wireless systems.

The results demonstrate that effective antenna performance can be achieved through intelligent geometrical design and ground-plane optimization, without relying on complex structures. This work contributes to the ongoing development of compact and efficient MIMO antennas for future wireless communication systems.

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